

Quality Choice with Reputation Effects: Evidence from Hospices in California

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Abstract

Hospices can build up their reputation by consistently choosing high quality, which is costly to provide. They are paid a fixed per-patient per-day rate by Medicare. I develop and estimate a dynamic oligopoly model of the California hospice industry to first show hospices compete on reputation, and then recover the hospice cost function. Counterfactuals show that i) higher reputation persistence leads to improved hospice quality, ii) lower Medicare reimbursement worsens hospice quality, and iii) partially tying Medicare reimbursement to quality lowers Medicare spending without sacrificing equilibrium quality.

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1 Introduction

Hospices provide palliative care to dying patients. Several policy concerns about the hospice industry motivate this paper. First, hospice care is a low-cost alternative that can significantly reduce end-of-life spending by moving many patients away from painful ineffective care. However, we need to incentivize good hospice quality for patients to switch; in turn, this requires us to understand how hospices choose quality of service.¹ Second, hospices have complained about low Medicare reimbursement rates. While this has reduced Medicare spending, is there any effect on hospice quality? Finally, Medicare reimburses hospices for each day a patient is enrolled. Can we improve upon this reimbursement scheme?

Using a dynamic oligopoly model of the California hospice industry, I show that hospices dynamically choose quality to build up reputation and attract consumers. This involves estimating consumer demand for hospices as well as the hospice cost function. Then, I use my model and estimates to evaluate counterfactual policies. Making reputation more persistent – such as having a hospice review website – improves hospice quality. Reducing Medicare reimbursement of hospices worsens hospice quality. Finally, partially tying reimbursement to quality lowers Medicare spending without worsening hospice quality.

The hospice industry is useful to study for several reasons. First, the US hospice industry is large and growing – over 1.6 million Medicare patients used hospices in 2022, and Medicare spent around \$21 billion on hospice services in 2019. However, very few papers empirically study incentives and regulation design for this industry. Second, it sheds light on policy design for “regulated price” healthcare settings where the government fixes reimbursement rates. Finally, it is a clean setting for studying how firm reputation affects consumer and firm choices, particularly in healthcare markets.

Hospice care is given to dying patients at their residences, where hospice staff administer pain medication and help with living arrangements. More visits by the hospice staff can be thought of as implying higher quality – it means the hospice is checking up on

¹Concerns have been raised about hospice quality across the US. The Department of Health and Human Services has released a report citing numerous deficiencies in hospice quality: <https://oig.hhs.gov/newsroom/media-materials/media-materials-2019-hospice/>

patients more and adjusting for day-to-day difficulties. Therefore, my measure of hospice quality is the average visits-per-patient-day. There is no price competition because most hospice patients are covered by Medicare, and Medicare pays hospices a fixed per-day rate for each patient. This makes hospice reputation particularly salient for consumers. Hospices which have given higher quality care in the past are more likely to be known and referred within the community. This suggests that a hospice can build up its reputation by persistently choosing high quality over time, and use this improved reputation to attract more patients.

To study reputation and quality choice in this setting, I build a structural model of hospice demand by consumers and dynamic quality choice by hospices. Reputation of a hospice is defined to be a stock of its current and past quality choices. This reputation stock partially depreciates when moving from one period to next. On the demand side, each consumer makes a discrete choice from a set of hospices in her market. Her choice is influenced by hospices' reputations and characteristics. On the supply side, hospices play a dynamic game in an oligopoly setting where they choose quality every period. They take into account that higher quality will lead to increased reputation and market share but also higher cost per consumer.

I estimate this model using annual hospice-level data from 28 counties in California for 2002-2018. The estimation proceeds as follows. First, I estimate a nested-logit discrete choice demand model following [Berry \(1994\)](#) and allowing for persistent unobserved demand shocks. This lets me pin down the magnitude and persistence of hospice reputation for consumer choice. Second, I regress hospice quality choices on estimated reputation stocks and other hospice characteristics to demonstrate what factors influence hospice quality choice. Following the arguments of [Berry and Compiani \(2023\)](#), I use hospice-specific fixed effects and instruments to account for serially correlated unobservables. Third, I estimate a dynamic oligopoly model of quality choice which incorporates reputation effects. This model describes how forward-looking hospices choose quality over time in a strategic setting. The estimation method follows [Bajari et al. \(2007\)](#). By incorporating additional data on hospice cost, this estimation procedure recovers the hospice cost function. The demand estimates show that reputation plays a significant role in consumer choice and depreciates at an annual rate of 46%. The hospice cost estimates

show that an additional visit by a hospice costs around \$192.

Using the estimated demand and cost parameters, I solve for hospices' quality choice probabilities via policy iteration. To illustrate quality choice dynamics, I plot the policy function against various state variables. Among others, I find evidence of reputation building and milking incentives. Then I solve for equilibrium quality choices under various counterfactual policies.

My first set of counterfactuals involves changing the persistence of hospice reputation. A policy that increases the persistence of hospice reputation – for instance, a review website that centralizes information on quality provision – leads to higher hospice quality. Such a review website called “Hospice Compare” was introduced by CMS in 2017, suggesting that CMS has reached a similar conclusion. The website has seen insufficient activity from consumers; hence, my counterfactual suggests incentivizing consumer participation on this website can promote quality competition with little additional cost to CMS.

My second set of counterfactuals involves changing Medicare's reimbursement rates for hospices. I find that lower (higher) Medicare rates leads to firms choosing lower (higher) quality. Higher rates can also incentivize entry by new hospices, and the resulting competition further improves hospice quality; correspondingly, lower rates can induce exit and worsen quality further. Therefore, CMS should be cognizant of the effect of lower reimbursement on quality of service. Higher quality from higher reimbursement can also cause patients to switch away from even more expensive care, such as that documented in [Gruber et al. \(2023\)](#), potentially generating net savings for Medicare.

My third set of counterfactuals involves changing the reimbursement scheme structure for Medicare. I compare the current per-day Medicare reimbursement scheme with a hybrid per-day per-visit reimbursement scheme. The latter is found to achieve the same equilibrium quality level but with lower Medicare spending. Therefore, partially tying reimbursement to quality can lower spending without harming quality.

Contributions and related literature: First, I contribute to a very sparse literature on the hospice industry and its regulations. I am the first to structurally estimate the supply-side of this industry. Prior work on the hospice industry include [Chung and Sorensen \(2018\)](#), who estimate a nested logit demand model to examine market expansion; [Dalton](#)

and Bradford (2019), who study length of stay of patients in non-profit and for-profit hospices; and Gruber et al. (2023), who show that hospices save money for Medicare by offsetting alternative expensive care for Alzheimer’s and Dementia patients.

Second, I am the first to build and estimate a structural dynamic oligopoly model of reputation accumulation through quality choice. I also construct a novel way of measuring reputation using market share data. This is in contrast with other papers that measure reputation with user reviews; my approach avoids the econometric issues involving user reviews and allows reputation to be studied in settings where such review data are not available. Very few papers attempt to analyze reputation acquisition via a structural model of firms. Apart from myself, two papers which do so are Saeedi (2019) and Bai (2024); my model and estimation are different from theirs. Saeedi (2019) studies how eBay sellers choose quantity over time to qualify for Powerseller status or Registered-Store status. In contrast to my paper, her model assumes quality of sellers is exogenous and buyers are unaware of past buyers’ experiences. Bai (2024) studies watermelon sellers using an experiment where some watermelons are laser-branded as a costly signal of quality. Unlike my paper, she does not assume that sellers can precisely choose their quality. Her demand model explicitly accounts for consumer learning and beliefs while I do not. In contrast to this paper, she does not estimate the supply side of her setting. There is also an extensive theoretical literature on reputation; for an overview see Bar-Isaac and Tadelis (2008).

Third, I show the importance of reputation for patients choosing healthcare providers as well as healthcare providers choosing quality. This phenomenon is understudied in the health economics literature and potentially extends to other industries, particularly those with non-contractible quality or where a firm’s brand capital plays a key role. There is a large empirical literature on reputation effects with most papers centered around online platforms, such as Cabral and Hortaçsu (2010), Fan et al. (2016), Newberry and Zhou (2019), and Tadelis (2016). There is much less work on reputation outside online platforms. One such paper is Jin and Leslie (2009) who study hygiene quality and reputation incentives in the context of restaurants; another is Bachmann et al. (2023) which studies spillover effects on reputation from the Volkswagen emissions scandal.

Fourth, I add to the literature on how healthcare providers choose quality. In contrast to other papers in this area, I show that quality choice has dynamic effects through

influencing provider reputation. Papers on healthcare provider quality include [Camarda \(2022\)](#), [Einav et al. \(2018\)](#), [Gaynor \(2006\)](#), [Gaynor et al. \(2013\)](#), [Grieco and Mcdevitt \(2017\)](#), [Hackmann \(2019\)](#), [Hotz and Xiao \(2011\)](#), [Lin \(2015\)](#), [Wang \(2022\)](#), and [Xiao \(2010\)](#). Similar to my paper, [Einav et al. \(2018\)](#) study alternative reimbursement schemes that can generate financial savings without harming quality, in their case for long-term care hospitals.

A strand of literature that this paper relates to is the advertising and brand management literature in marketing. Several structural papers in that literature look at how firms spend money on advertising to build up their brand equity; see [Dubé et al. \(2005\)](#) and [Borkovsky et al. \(2017\)](#), and the references therein. This paper also adds to the literature on estimating dynamic games. I use the framework from [Ericson and Pakes \(1995\)](#), and use estimation methods from [Bajari et al. \(2007\)](#) and [Pakes et al. \(2007\)](#). Other papers on similar methods include [Aguirregabiria and Mira \(2007\)](#), [Collard-Wexler \(2013\)](#), and [Ryan \(2012\)](#); for detailed overviews see [Doraszelski and Pakes \(2007\)](#), [Arcidiacono and Ellickson \(2011\)](#), and [Aguirregabiria et al. \(2021\)](#). My paper has similarities with [Benkard \(2004\)](#), in that I use a nested-logit demand framework within a dynamic game of accumulation. While my paper is about reputation accumulation, [Benkard \(2004\)](#) is about accumulating experience levels. This paper also relates to the dynamic investment literature in industrial organization; classic references are [Olley and Pakes \(1996\)](#) and [Ericson and Pakes \(1995\)](#).

2 Industry background

Hospices provide palliative care to terminally ill patients, the majority of whom die within 1 week to 3 months in my dataset. Patients who enroll in hospice typically no longer receive curative treatment. Most patients in the dataset are covered by Medicare’s reimbursement scheme, which pays hospices a fixed amount for every day the patient is enrolled. In return, hospice staff provide care to the patient by visiting her at her residence. The number of visits a hospice makes to its patient is a strong measure of the quality of service, and I claim that a hospice can build up its reputation by consistently providing high quality over time.

2.1 Hospice care provision

Unlike other medical providers, hospices typically provide care at the residence of the patient. Nurses and home health aides make the majority of hospice visits.² The hospice staff make multiple visits to a patient throughout the patient's enrollment period. Administrators of a hospice decide the number of visits, and set up the schedule for each nurse. A day for a hospice nurse involves driving to several patients as determined by the administrator, giving them care, writing up reports for the hospice physician and administrator, and refilling supplies.

The care given by hospice nurses is relatively low-skill compared to other medical providers. Hospices are not responsible for curative treatment; they are tasked with pain management and ease-of-living for a dying patient. Examples of hospice care include administering pain medication, providing medical supplies like oxygen and bandages, dressing bedsores, giving physical and speech therapy, helping with bathing and feeding, temporarily substituting as the primary caregiver, tending to emotional and spiritual needs of the patient, and giving grief counselling to the family after the patient passes away.³

It is important to note that the hospice decides how many visits to make to each patient and what care is to be provided during each visit. As is explained below, the patient does not incur any additional cost for more visits, and so would plausibly like as many visits as possible.

2.2 Hospice selection by patients

How a hospice is chosen by a patient and her family vary. Sometimes a social worker at the hospital gives the patient a list of hospices, and suggestions on which to choose based on past experience; the same can be done by the patient's physician. The family

²Hospices employ several types of staff: registered nurses, licensed vocational nurses, home healthcare aides, physicians, social workers and chaplains. Physicians rarely make visits, and usually deal with reports on the patients from nurses. Social workers help the patient and her family with paperwork and choices.

³Hospice staff do not function as the primary caregiver of the patient - someone in the patient's residence (a family member if the patient resides at home, or a nurse if the patient resides in a nursing facility) has to be the primary caregiver. But hospice nurses assist the primary caregiver, and sometimes may substitute in for a period of time to give respite to the primary caregiver.

can also search for hospices online (aided by online reviews and forums), or can rely on word-of-mouth from other families that have needed hospice services in the past.

When looking for advice on how to choose a hospice, a few suggestions are common. One is to choose hospices that are reputable and have served the community well over a long time.⁴ Another is to choose a hospice that makes regular visits, is readily available for emergencies, and is ready to give care to patients with various complexities and circumstances. Additional suggestions include asking how the hospice will co-manage with and give respite to the primary caregiver. These suggest the following:

1. Reputation matters for hospice choice. A hospice which has provided high quality of care in the past is more likely to be suggested by social workers, better known in the community and better reviewed than a hospice which has not. Such a hospice is more likely to be selected by a patient and her family, and so will have greater market share. Presence of intermediaries such as social workers also suggests how information about a hospice's past quality choices can persist over time.
2. Quality of care is strongly dependent on the number of visits that a hospice makes to a patient. A hospice which makes more visits is constantly checking and adjusting to the patient's condition, can manage symptoms better, and is likely to arrive quickly in case of an emergency. Such a hospice is also able to provide more respite to the patient's primary caregiver, which may be highly valued by the family. Importantly, it is also likely that number of visits is correlated with the general effort of the hospice, and so with other unobserved dimensions of quality.⁵ This is because visits

⁴Some websites which give suggestions on how hospices are chosen are listed in Appendix G.2 alongside relevant quotes. A representative quote from AmericanHospice.org reads:

"What do others say about this hospice? Get references both from people you know and from people in the field – e.g., local hospitals, nursing homes, clinicians. Ask anyone that you have connections to if they have had experience with the hospice and what their impressions are. Geriatric care managers can be a particularly good resource, as they often make referrals to hospices and hear from families about the care that was provided."

Similarly, HospiceFoundation.org states:

"Seek professional opinions. Ask clinicians, professional caregivers at nursing homes, geriatric care managers, or end-of-life doulas about their experience with a hospice. Talk to friends, family, and neighbors who have used hospice services and get their opinions about the experience with a provider."

⁵CMS and AHRQ have, in recent years, started tracking a variety of hospice service measures; see Appendix G.1 for details.

are the most costly part of care - the hospice has to hire extra nurses and pay the fuel cost of traveling. If a hospice is incurring this cost already, it is unlikely that they are then shirking on within-visit care.⁶ That is, visits should be a good proxy for the overall effort that a hospice exerts.

While the above suggest that reputation and number of visits are important determinants of hospice choice, I estimate a demand model to check if that is indeed the case.

2.3 Payers and reimbursement schemes

For hospice reimbursement, I focus my analysis on eligibility rules and reimbursement schemes for Medicare only. While hospice services can be paid for by a variety of payers – Medicare, Medi-Cal (California’s Medicaid), private insurance, charity, self-pay, etc. – the overwhelming majority of patients are paid for by Medicare (see Table 1). Furthermore, Medi-Cal’s reimbursement is essentially the same as Medicare’s.

For a patient covered by Medicare, hospice care is essentially free. There is no out-of-pocket cost for enrolling into or receiving visits from a hospice. Patients have to give a copay of up to \$5 for outpatient drugs required for pain management. This means that the “price” of choosing a hospice does not vary across hospices in the patient’s choice set, and moreover can be thought of as zero. The patient generally has to turn down curative treatment to enroll into hospice care via Medicare.

For the hospice, enrolling a patient covered by Medicare results in the hospice getting paid a fixed rate for every day the patient is enrolled (i.e. a per-diem rate). It is important to note that the fixed rate is set exogenously and does not depend on the number of visits that the hospice staff makes. To contrast extreme examples, if a patient enrolls in a hospice for 10 days, the hospice gets paid a fixed amount for each of those 10 days irrespective of

⁶Hospice care mostly involves relatively low-skill care such as giving pain medication, making adjustments to living arrangements, and helping out the patient with certain physical activities. As such, it is unlikely that there will be much heterogeneity across hospices regarding what is done during visits. Furthermore, in my demand model, I allow for persistent unobserved hospice quality.

how many visits they make.⁷⁸

The Medicare rate is determined as follows. Medicare sets a national per-diem rate that is split into a “labor component” and a “non labor component”. Medicare also creates wage indices for each Metropolitan Statistical Area (MSA), such that a MSA with high wages will have a high wage index. The per-diem reimbursement rate of a hospice is the non-labor component and the labor component times the wage index of the hospice’s MSA.⁹ These rates are updated every year.

Payer	% of patients
Medicare	83.7
Medi-Cal	7.37
Private insurance	6.37
Selfpay	1.65
Charity	0.82

Table 1: Percentage of patients in dataset covered by each payer type.

⁷Medicare also distinguishes between different types of hospice care. Specifically, it draws a distinction between routine care (which is the care described above), inpatient care (where a patient moves to an inpatient ward of a hospice for intense treatment) and continuous respite care (where the hospice takes over as the primary caregiver). The majority of patients in my sample are given routine care; I show in Appendix B that of all the “days of care” provided by a hospice within each year, over 99% is dedicated to routine care. Medicare allows the per-diem rate to vary substantially based on whether the care provided is routine home care, inpatient care, or continuous care.

⁸There are additional dimensions of the Medicare reimbursement scheme that I do not study in this paper. First, the total revenue paid by Medicare to a hospice is subject to a cap that rises with the number of patients enrolled in the hospice. For a detailed study, see Gruber et al. (2023). Second, from 2017, Medicare began paying hospices an additional amount for visiting a patient in the last week of her life (called the “service intensity add-on”). Third, Medicare also lowers the per-diem rate for routine care if a patient is enrolled for more than 60 days to discourage hospices from enrolling non-terminal patients. Interestingly, it seems from news reports in 2019 that the service intensity add-on had seen limited uptake from hospices due to lack of understanding; see <https://hospicenews.com/2019/05/20/medicare-service-intensity-add-on-underused-by-hospices/>.

⁹To work out an example: for FY 2021, the labor component of routine home care is \$136.9, and the non-labor component is \$62.35. Suppose a hospice is in a county with a wage index of 0.8. The reimbursement rate of the hospice is approximately $136.9 \times 0.8 + 62.35 = \171.87 . See Appendix G.3 for an example of Medicare’s reimbursement rates and its variation across counties.

3 Data

The dataset for this paper comes from firm-level data on hospices in California for 2002-2018. I use these data to construct firm-level measures of quality, defined to be visits-per-patient-day made by a hospice in a year. A market is defined to be at the level of a county, and I restrict my estimation to 28 counties. Total firms in a market range from 1 to 23, with moderate entry and little exit.

3.1 Data sources

The main dataset comes from Home Health Agencies And Hospice Annual Utilization Reports compiled by California Department of Health Care Access and Information (CHAI). Hospices in California are required to submit yearly utilization reports to CHAI where they report firm-level statistics for the year. This includes measures such as current location, total number of patients, total visits per year (broken down by type of staff), hospice characteristics, aggregate characteristics of patient pool, annual expenses, etc. I collect and clean utilization reports of hospices for 2002-2018. I complement this with additional data sources on population sizes (by age), mortality rates, and Medicare hospice reimbursement rates for California over the same time period.

Additional details on how I deal with nonresponses, outliers, and firm IDs are specified in Appendix [D](#).

3.2 Market definition

A market is defined to be at the level of a county.¹⁰ Market size is constructed by multiplying the population size of each age bracket for a market with the corresponding mortality

¹⁰While some hospices offer services to multiple counties, I can rule out clusters of counties as a market definition using data. Specifically, the dataset contains a breakdown of a hospice's patient pool by patients' counties of residence. Most of a hospice's patients originate from the county where the hospice is located; see Appendix [B.4](#). This is reasonable since both the hospice and the patient have a strong preference to minimize distance – the patient wants to be close to the hospice so that she can benefit from quick visits during an emergency, and the hospice prefers serving nearby patients since it is less costly to travel to their residences. The only exception are Yuba and Sutter counties which have a large overlap of patients, and so I combine them into a single market.

rate.

I focus on 28 counties for my project; a list of these counties can be found in Appendix B.4. While California has 58 counties, I have to drop the remainder for the following reasons. First, several counties do not see any hospice presence over the course of my dataset. Second, a few counties have a very large number of hospices (Los Angeles county has over 300). Markets with many small sellers have very different dynamics than those with few sellers. My model and estimation are oriented towards strategic interactions among small numbers of players and does not naturally accommodate such active markets. Having few firms in a market also allows me to disentangle the competitive effects of rivals very clearly. Such sample selection is in line with many dynamic oligopoly papers such as Collard-Wexler (2013) and Lin (2015).

3.3 Measure of hospice quality

Following the discussion in Section 2.2, my measure of quality should reflect how often a hospice visits its patients. Recall that I have firm-level data; the dataset gives me total visits that a hospice makes to its entire pool of patients within a year and the number of patients it serves in the same year. The data also contains a breakdown of length-of-stay for the patient pool in a hospice-year (how many patients stayed for 0-7 days, 8-30 days, 31-90 days, 91-179 days, and 180+ days). This allows me to construct an “average length of stay” for patients at a given hospice-year. Table 2 shows the distribution of these variables.

I combine the above variables to construct my measure of quality “visits-per-patient-day” as follows. Multiplying the total patients at a hospice with its average length-of-stay gives the total patient-days at the hospice. Dividing total visits¹¹ in a year by a hospice

¹¹I aggregate all visits across staff type rather than select specific staff types, or allow different staff to contribute differently to the overall quality measure. This is because in my dataset, I see all staff types co-move very closely for hospices; see Section B.5 for Principal Components Analysis.

with its total patient-days gives the visits-per-patient-day.¹²

Variable	10%	25%	50%	75%	90%
Visits per patient	21.0	26.67	33.68	44.84	61.43
Average length-of-stay	35.91	42.28	49.82	58.19	66.54
Visits per patient-day	0.48	0.57	0.69	0.84	1.06
Patients	63.0	149.0	289.5	511.75	765.0
Market share	2.77	7.37	19.71	40.04	100.0

Table 2: Distribution of visits, average length-of-stay per hospice-year, visits-per-patient-day, patients, and within-hospice market share. Each observation is at the hospice-year level.

3.4 Summary statistics on competition and patients

I now present some summary statistics on competition and patient composition within my sample. The tables below and in Appendix B show distributions of various key variables; the main takeaways are as follows. First, the vast majority of hospices have over 100 patients. This assuages worries that my key measure of quality - visits per patient day - is mostly driven by fluctuating patient severity, since we would expect that variations in patient severity would even out over a large number of patients over the course of the year. Second, the median firm count is 2 with the 75th percentile at 4, suggesting that an oligopoly setting is the correct mode of analysis instead of perfect competition. Third, while the total entry and exit over the 16 years of my data is significant, there is very little turnover in a market within an individual year. In my model, I will treat exit as effectively exogenous due to the lack of variation in my data. Fourth, the vast majority of patients have a length of stay between a week and 3 months; see Appendix B for more details. Finally,

¹²It is often the case that patients need more visits near the end of their length of stay compared to earlier. Thus I construct the average length of stay to reflect this possibility. For the length-of-stay bins, I construct the chosen point in each bin to be 7, 25, 60, 100, and 230. Then the number of patients in each bin is multiplied with the corresponding point. The intuition is that patients in bins 0-7 days and 8-30 days need visits over most of their length of stay; however, longer staying patients may need visits for fewer days than their length of stay. I also estimate the demand system with visits-per-patient as the quality variable and obtain similar results.

Appendix B also does a breakdown of patients by diagnosis, revealing that roughly 40% of the patients are suffering from cancer.

Table 3: Distribution of market-level characteristics.

Variable	Min	10%	25%	50%	75%	90%	Max
Firm count	1.0	1.0	1.0	2.0	4.0	7.0	23.0
Entry count	0.0	0.0	0.0	0.0	0.0	1.0	5.0
Exit count	0.0	0.0	0.0	0.0	0.0	1.0	3.0
Medicare rates	134.48	151.88	178.29	212.41	255.25	284.52	375.92

Notes: Each observation is at the county-year level. Medicare per-day rates are inflation-adjusted.

The dataset contains information on hospice-specific services and characteristics. To give a snapshot, in 2017 there were 171 hospices in my dataset; of these, 12 had an inpatient unit, 15 offered special pediatric services, 2 offered adult day care services, and 95 were for-profit. The dataset also breaks down hospices by agency type: 110 hospices were free-standing, 27 also had a home-health unit, and 31 were hospital-based. I control for these characteristics in my demand estimation.

3.5 Cost measures from hospice cost data

The data includes breakdown of annual hospice expenses. From this, for each hospice-year, I construct measures of per-period fixed cost, per-patient non-visit cost (hereafter referred to as “intercept cost”), and cost of making an additional visit (hereafter referred to as “slope cost”). All cost measures used in the analysis are inflation-adjusted. Details of variable construction, as well as a list of all the variables in the cost data, are listed in Appendix H.

4 Structural model

To study and model reputation in this setting, I set up a structural model of demand and supply for the hospice industry. The demand side is a discrete-choice model that incorporates a measure of reputation of the hospice. The supply side is a dynamic oligopoly

model where forward-looking hospices are choosing quality in a competitive setting. The remainder of the section details the equilibrium concepts and the timing of the dynamic game.

4.1 Demand

The demand for hospices is modeled as a nested-logit discrete choice model, with a nest on hospices. A consumer can choose between any hospice present in the market in that period, or an outside option. The outside option of not choosing a hospice can be interpreted as the consumer continuing curative treatment, or choosing to be cared for by family/nursing home staff.

The utility of consumer i from choosing hospice j in period t is given by:

$$u_{ijt} = \alpha_{m(j)} + X'_{jt}\beta_d + \psi_{jt} + \xi_{jt} + \zeta_i + (1 - \sigma_n)\tilde{\epsilon}_{ijt} \quad (1)$$

$$\xi_{jt} = \rho\xi_{jt-1} + \varepsilon_{jt} \quad (2)$$

where $m(j)$ denotes the market in which hospice j is present and X_{jt} represents hospice characteristics¹³. The term ξ_{jt} denotes unobserved hospice-specific quality that is allowed to be serially correlated and vary over time. This can reflect factors like how connected the hospice is with physicians/nursing homes, advertising, and care quality beyond number of visits. The term ψ_{jt} is the reputation stock of hospice j in time t . Finally, $\zeta_i + (1 - \sigma_n)\tilde{\epsilon}_{ijt}$ reflect that this is a nested logit - there are two nests, one on all hospices and one on the outside option.

Reputation is modeled as a stock transition equation:

$$\psi_{jt} = (1 - \tau)\psi_{jt-1} + \eta a_{jt} \quad (3)$$

where a_{jt} is the visits-per-patient made by hospice j in period t . Assuming a new entrant

¹³The hospice characteristics are dummy variables for the following: hospice inpatient unit, pediatric program, bereavement services, day care for adults, for-profit status, free-standing hospice, home-health-based hospice, and hospital-based hospice.

has a reputation of zero, this can be rewritten as:

$$\psi_{jt} = \eta[a_{jt} + (1 - \tau)a_{jt-1} + (1 - \tau)^2 a_{jt-2} + \dots]$$

That is, a firm's reputation is a discounted sum of its current and past choices of quality.¹⁴ I now offer a more thorough discussion on the modeling choices.

First, the nest on hospices (i.e. the inside options) allows for more flexible substitution patterns between hospices and the outside option relative to a simple logit with no nests. For instance, a high value of the nest parameter implies that should a consumer decide not to pick hospice j_1 , she is more likely to pick another hospice j_2 instead of switching to the outside option; i.e. a consumer's preferences are positively correlated within the nest.

Second, the persistence of unobserved hospice-specific quality ξ_{jt} allows me to capture persistent unobservables in demand. One example could be that a hospice is well-connected with physicians and social workers in some hospitals and so get repeated referrals despite not choosing higher quality - this will manifest as the hospice having high ξ_{jt} for many periods. Another example could be a hospice that has suffered a recent scandal such as elder abuse. Despite choosing high visits-per-patient-day the negative publicity may last for several periods - this will manifest itself as the hospice having low ξ_{jt} for many periods. Importantly, this allows for the fact that a hospice with persistent demand shocks will adjust its quality choices accordingly; for example, a hospice that predicts getting high draws of ξ_{jt} in the future may choose to lower its quality level in the current period. In addition, this allows for the possibility that a hospice with high ξ might face less entry because it is more difficult to compete against. In the demand estimation this means I build moments around ε_{jt} instead of ξ_{jt} ; this follows [Grennan \(2013\)](#) and [Lee \(2013\)](#). The reasons behind why this is a more reasonable exclusion restriction are explained in [Section 6.1](#).

¹⁴Note that the impact of a firms' age is already subsumed in my reputation transition equation (a firm who has lived longer has accumulated a larger number of past quality choices into its current reputation stock).

4.2 Supply

To understand how hospices choose quality over time, I set up a discrete-time infinite-horizon dynamic oligopoly model. To do so, I combine the above demand system with cost function and payoff function to formulate a strategic dynamic optimization problem. This broadly follows the framework of [Ericson and Pakes \(1995\)](#). In subsequent sections, I lay out i) cost function, ii) per-period profit function, iii) dynamic programming problem, iv) equilibrium concept, and v) model-predicted choice probabilities.

In the dynamic game a period is defined to be a year. For the rest of this section the time subscript is suppressed for clarity unless necessary.

4.2.1 Cost function and per-period profit

The cost to a hospice for serving a patient depends on its quality choice and other cost components. I impose the cost to be linear in visits-per-patient-day.

The expected cost of serving each patient at quality a_j (henceforth referred to as marginal cost) is given by¹⁵:

$$MC_j(a_j) = \gamma_{0j} + \gamma_{1j}a_j\overline{LOS} \quad (4)$$

where $\overline{LOS}$ is the expected length-of-stay for a patient. Thus $a_j\overline{LOS}$ captures the average number of visits made to a patient over her entire length of stay, γ_{1j} is the cost of an additional visit, and γ_{0j} captures the non-visit cost for each patient. The functional form captures the intuition that marginal cost is linearly increasing in quality a_j given $\gamma_{1j} > 0$. From before, I refer to γ_{1j} as the *slope cost* and γ_{0j} as the *intercept cost*.

Combining the demand function with Medicare prices and market size, the per-period profit of hospice j in market m choosing quality a_j is:

$$\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) [P_m^{MCAR} \overline{LOS} - MC_j(a_j)] - \phi_j \quad (5)$$

¹⁵Given the richness and panel structure of my cost data, I can allow cost parameters (slope cost, intercept cost, per-period cost) to vary across firms *and* over time. However for estimation, I impose additional restrictions.

where $\mathbf{x}_m$ denotes the state variables in market m , $\mathbf{a}_{-j}$ denotes the vector of realized actions of all rival firms in the market, M_m is the market size, and ϕ_j is the per-period fixed cost of hospice j . Medicare's per-day rate P^{MCAR} is multiplied with the expected length-of-stay $\overline{LOS}$, giving the expected revenue generated by a patient over their length-of-stay. This profit function is interpreted as maximizing the sum of *expected* profit per patient. This is because when choosing quality, a hospice does not know the exact realizations of length-of-stay for each patient it will enroll. In Appendix F.2 I derive this profit function and discuss alternatives.

For the demand estimation I use the continuous measure of quality choice; however for estimating the supply side, I discretize quality choices into six tiers. This has the advantage of improving computational tractability (especially regarding counterfactuals) while still being a meaningful improvement of quality.

4.2.2 Dynamic choice of quality

The dynamic oligopoly game is modeled as a discrete-time simultaneous-move game over an infinite horizon. Every period (year) the incumbents decide what level of quality to provide. All consumers of a hospice receive the quality level chosen. Persistence of reputation means that current quality choices affect future sales, hence a forward-looking model is required; hospices also make this decision in a competitive setting, hence a model of strategic interaction is needed.

Hospice j 's value function is:

$$V_j(\mathbf{x}_m, \boldsymbol{\varepsilon}_j^a; \boldsymbol{\sigma}; \boldsymbol{\theta}) = \max_{a_j \in \mathcal{A}} \mathbb{E} \left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \boldsymbol{\theta}) + \varepsilon_j^a(a_j) + \beta V_j(\mathbf{x}'_m, \boldsymbol{\varepsilon}'_j{}^a, \boldsymbol{\sigma}; \boldsymbol{\theta}) \middle| a_j, \mathbf{x}_m, \boldsymbol{\sigma} \right] \quad (6)$$

where $\boldsymbol{\theta}$ is the set of structural supply-side parameters, $\beta \in (0, 1)$ is the per-period discount rate, $\boldsymbol{\varepsilon}_j^a$ is a vector of choice-specific errors for hospice j , $\varepsilon_j^a(a_j)$ is the choice-specific errors for hospice j choosing action a_j , and $\boldsymbol{\sigma}$ is the equilibrium strategy profile of the hospice and its rivals (to be defined later). The symbol $\mathcal{A}$ denotes the set of possible quality choices (recall that we discretize quality into six tiers for the supply side estimation), and a_j is the quality level chosen from this set. The choice-specific error terms are meant

to capture unobservables affecting a firm's quality choice. This choice-specific structural error term is known to the firm but unknown to the econometrician. In keeping with the literature and to help with tractability in estimation, I assume these shocks to be *i.i.d* and distributed Type-1 Extreme Value.

The above setup reflects two assumptions about the choice-specific shocks, following Rust (1987). First, the choice-specific shocks enter the payoff function additively. Second, the choice-specific shocks evolve independently of other state variables.¹⁶

4.2.3 Equilibrium and Conditional Choice Probabilities

This section describes the equilibrium concept used and how it leads to the conditional choice probabilities (CCPs) predicted by the model. These CCPs will later be matched to observed probabilities to estimate the cost parameters. For this subsection, the notation for parameters θ is suppressed from the arguments for clarity.

In keeping with the literature on empirical dynamic games, I focus on pure-strategy symmetric anonymous Markov Perfect Equilibrium (MPE). Each firm's strategy thus depends only on current period's payoff-relevant state variables and its private choice-specific shocks; see Appendix E.2 for further explanation. Therefore a firm's strategy can be written as a mapping from current-period states and private choice-specific shocks to actions:

¹⁶Below I list these assumptions explicitly; the following mirrors the discussion in Lin (2015). Let the vector of common knowledge state variables in the current period be $\mathbf{x}$. The vector of common knowledge state variables and private information is denoted as $\mathbf{s} = (\mathbf{x}, \varepsilon^a)$. The state-to-state transition probability is $F(\mathbf{s}'|\mathbf{s}, \mathbf{a})$, where $\mathbf{a}$ denotes the vector of firm choices in the current period. Following Rust (1987), the two assumptions can be stated as:

1. Additive Separability: This imposes that the choice-specific structural error ε^a enters the per-period payoff function additively, i.e. $\pi(a_j, \mathbf{a}_{-j}, \mathbf{x}_m, \varepsilon_j^a; \theta) = \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) + \varepsilon_j^a(a_j)$.
2. Conditional Independence: This comprises two assumptions. One, $\mathbf{x}'$ is independent of ε^a after conditioning on $(\mathbf{x}, \mathbf{a})$. Two, ε^a is *i.i.d.* and evolves independently of other state variables. Mathematically, this can be written as:

$$F(\mathbf{s}'|\mathbf{s}, \mathbf{a}) = F(\mathbf{x}', \varepsilon^a|\mathbf{x}, \varepsilon^a, \mathbf{a}) = F(\mathbf{x}'|\mathbf{x}, \mathbf{a})F_\varepsilon(\varepsilon^a)$$

where $F_\varepsilon(\cdot)$ denotes the distribution of ε^a .

$$\sigma_j : (\mathbf{x}_m, \varepsilon_j^a) \rightarrow \mathcal{A}$$

Let the vector $\sigma = \{\sigma_j\}_{\forall j}$ denote the strategy profile of all firms in a market. I now show how this can be used to construct CCPs.

The *choice-specific value function* of action a in state $\mathbf{x}_m$, $\hat{v}_j(a, \mathbf{x}_m, \sigma)$, is the discounted sum of payoffs to hospice j from choosing action a before choice-specific shocks ε_j^a are observed. Let the *ex-ante value functions* $V_j(\mathbf{x}_m, \sigma)$ be Equation (6) with the choice-specific errors integrated out:

$$V_j(\mathbf{x}_m, \sigma) \equiv \int V_j(\mathbf{x}_m, \sigma, \varepsilon_j^a) dG(\varepsilon_j^a) \quad (7)$$

The ex-ante value function can be written in terms of choice-specific value functions:

$$\begin{aligned} V_j(\mathbf{x}_m, \sigma) &= \int V_j(\mathbf{x}_m, \sigma, \varepsilon_j^a) dG(\varepsilon_j^a) \\ &= \int \max_{a_j \in \mathcal{A}} \{ \hat{v}_j(a_j, \mathbf{x}_m, \sigma) + \varepsilon_j^a(a_j) \} dG(\varepsilon_j^a) \end{aligned} \quad (8)$$

The MPE is the strategy profile σ^* such that every firm is choosing the optimal strategy given the strategies of their rivals:

$$\sigma_j^*(\mathbf{x}_m, \varepsilon_j^a, \sigma^*) = \arg \max_{a_j \in \mathcal{A}} \{ \hat{v}_j(a_j, \mathbf{x}_m, \sigma^*) + \varepsilon_j^a(a_j) \}$$

Given this optimal strategy profile, the CCP of an action a_j taken by firm j can be written as:

$$\Psi_j(a_j, \mathbf{x}_m, \sigma^*) = \frac{\exp \{ \hat{v}_j(a_j, \mathbf{x}_m, \sigma^*) / \theta_e \}}{\sum_{a \in \mathcal{A}} \exp \{ \hat{v}_j(a, \mathbf{x}_m, \sigma^*) / \theta_e \}} \quad (9)$$

where θ_e is the logit scaling parameter of the Type-1 Extreme Value logit error ε^a . Since the choice-specific structural error is known to the firm but unknown to the researcher, I integrate out the errors and recover the CCP. The CCP therefore gives us the choice probabilities of all available actions for a given observed state.

Under the assumption of logit error, we can write the choice-specific value function

$\hat{v}(a_j, \mathbf{x}_m, \sigma^*)$ as¹⁷:

$$\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*) = \sum_{\mathbf{a}_{-j} \in \mathcal{A}_{-j}} \left\{ \left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) + \beta V_j(\mathbf{x}'_m, \sigma^*) \right] F(\mathbf{x}'_m | \mathbf{x}_m, a_j, \mathbf{a}_{-j}) \prod_{n \neq j} \Psi_n(\mathbf{a}_{-j}[n], \mathbf{x}_m, \sigma^*) \right\} \quad (10)$$

where $\Psi_n(\cdot, \mathbf{x}_m, \sigma^*)$ is the CCP of firm n , $\mathcal{A}_{-j}$ is the set of all possible rival action profile vectors, and $\mathbf{a}_{-j}[n]$ is the action by firm n .

The logit assumption also allows us to rewrite Equation (8) as:

$$V_j(\mathbf{x}_m, \sigma^*) = \theta_e \left[0.577216 + \ln \left(\sum_{a_j \in \mathcal{A}} e^{\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*) / \theta_e} \right) \right] \quad (11)$$

4.3 Timing

In any period, the industry state comprises common-knowledge variables (own characteristics, rival characteristics, market characteristics) and private information (choice-specific structural errors ε^a). The industry state comprises both endogenous and exogenous variables, and as a result evolves due to a mix of firm decisions and exogenous transitions.

Given the model above, the game's evolution and timing is as follows. For period t and market m :

1. Incumbents observe $\mathbf{x}_{mt}$ and structural errors, and each make quality choices.
2. Reputation stock of each incumbent evolves.
3. Consumers observe $\mathbf{x}_{mt}$, reputation stocks, and structural errors, then choose a hospice.
4. Market shares are realized, and incumbents receive profits.

¹⁷Prior to knowing the choice-specific structural errors, $\hat{v}(a_j, \mathbf{x}, \sigma^*)$ is hospice j 's expected discounted sum of payoffs from choosing action a_j when facing state variables $\mathbf{x}$. Forming this expectation requires integrating over several random variables. The last term shows that this expression requires integrating over rival actions, which is done using the CCP at equilibrium strategies. Intuitively, since this is a simultaneous-move game, when hospice j chooses an action this period it does not know its rivals actions, and so has to make predictions using the equilibrium strategies. The middle term shows that conditional on own choice and rival choices, the firm also needs to integrate over all possible values of state variables. The first term is then the current period payoff and continuation value conditional on state variables and rival choices.

5. Incumbents stay or exit the market.
6. Potential entrants observe $\mathbf{x}_{mt}$, then decide whether to enter or disappear.
7. All state variables evolve.

5 Discussion

I now offer a more thorough discussion on how reputation operates in my setting and explain the motivations behind the reputation transition equation (Equation (3)).¹⁸

In my model consumers care about a hospice’s reputation. Holding everything constant, a consumer is more likely to choose a hospice with high reputation over one with low.¹⁹ Given this setting, why do consumers care about a hospice’s reputation? A few reasons are:

1. A hospice with higher reputation has more goodwill and recognition in the community, and so is more likely to be referred to the patient. This has parallels with the intangible brand capital literature - for example [Bronnenberg et al. \(2022\)](#) talks at length about brand equity and reputation, and how these create an incentive for firms to repeatedly supply high quality over time. It also has parallels with reputation in a repeated game of quality choice under complete information.
2. To predict quality choice, consumers can track a firm’s reputation alone instead of all possible state variables. As can be seen in the model above, quality choice is affected

¹⁸Note that instead of modeling reputation formation as a complex process of belief updating, I approximate the reputation stock of a hospice as a parametric function of its current and past quality choices. This reputation stock enters a consumer’s utility function and influences her choice of a hospice. This parametric approximation allows me to study reputation using only firm-level data and maintain computational tractability. It also allows me to be agnostic about the specific channel through which reputation operates. Consumers may care about reputation because of the corresponding goodwill and name recognition, or because reputation is a good statistic for predicting future quality in the face of bounded rationality.

¹⁹Using the terminology from the theoretical literature on reputation, my model is somewhat similar to a hidden action model of information (as opposed to hidden information). Hidden action models assume that quality is chosen by the firm and consumers try to predict a firm’s quality choice with full information about its past choices. In contrast, a hidden information model assumes that a firm’s quality is exogenously fixed, and the firm can choose to signal this quality type to consumers.

by factors such as costs, unobserved demand shocks, and hospice characteristics, all of which are serially correlated. Instead of keeping track of all these variables and their evolution, consumers can use reputation as a statistic for predicting the quality they will receive from the hospice. This reputation index can be learned by talking to social workers, physicians, and other information sources.

Another element of the model is that reputation depreciates at the rate of τ . Why does reputation depreciate? There are several possibilities. One, information on past quality choices disappears over time. This might be because of forgetting by the community. It could also be due to retirement or relocation of physicians, social workers, and community members. The patient's family is less likely to require a hospice soon after the patient passes away, so the fact that there are no repeat consumers means information is less likely to persist. Another way of thinking about this is that information on past choices is still available but gets more costly to retrieve over time. Two, past quality choices may be less relevant over time. This is because past quality choice also represents old and outdated medical practice styles, technology, and medicine. An example is that a hospice nurse's visit 20 years ago had very different content compared to a hospice nurse's visit at present. Third, quality choices very far in the past reflect the hospice's demand shocks, costs, and characteristics that may no longer have any predictive power, since these are also changing over time.

Finally, note that my reputation transition equation includes current quality as well as past quality. This means that current period reputation – which influences hospice choice by a consumer – is affected by current quality choice. This seems to indicate that consumers observe current quality choice; if so, why would they need to predict current quality choice using past choices? First, the decision to include current quality into reputation is mainly driven by the fact that my data is at the yearly level. This means information on current quality choice has time to diffuse through the community over the course of a year. Second, it subsumes models of quality competition in most empirical papers that include only current period quality in the demand system (e.g. [Hackmann \(2019\)](#), [Lin \(2015\)](#)). Finally, it accounts for the possibility that a hospice might be able (to some extent) to convince a potential consumer about the level of quality they will provide.

See Appendix I for additional discussions on institutional details and modeling decisions, such as advertising, selection, capacity constraints, etc.

6 Estimation and Results

This section details how the structural model above is estimated and the estimation results. The demand model is estimated using two-stage GMM following the inversion technique of Berry (1994). The supply side is estimated by leveraging additional cost data and using the method of Bajari et al. (2007) that matches simulated choices with observed choices. Demand estimates show that consumers are more likely to choose a hospice with higher reputation, and the reputation stock depreciates by 46% every year. Supply side estimates show that an additional visit by a hospice to a patient costs around \$192. I also regress observed quality on estimated reputation stocks and characteristics of a hospice and its rivals to further understand hospice quality choice. Additional details about the estimation procedures are given in Appendix E.

6.1 Demand estimation

Using Berry (1994), the nested-logit discrete choice model can be rewritten in terms of observed market shares:

$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n \ln(s_{j|gt}) + \xi_{jt} + \eta[a_{jt} + (1 - \tau)a_{jt-1} + (1 - \tau)^2 a_{jt-2} + \dots] \quad (12)$$

where s_{jt} and s_{0t} represent the market shares of hospice j and the outside option respectively, and $s_{j|gt}$ is j 's within-hospice market share. Recall that for flexibility we also allow the unobserved demand shocks to be persistent:

$$\xi_{jt} = \rho \xi_{jt-1} + \varepsilon_{jt}$$

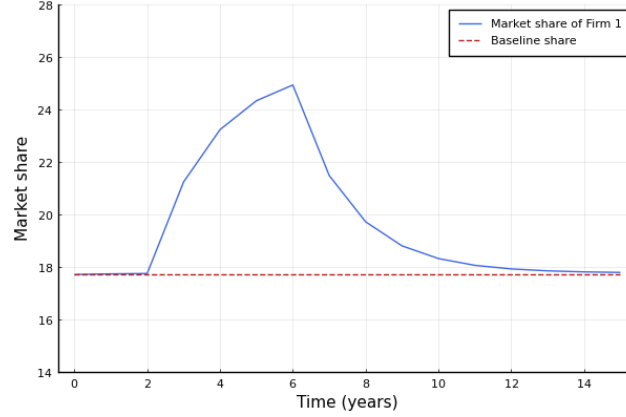
These equations are estimated simultaneously via two-step GMM. Endogeneity is an issue since the within-hospice market share $s_{j|gt}$ is mechanically correlated with unobserved demand shock ξ . Furthermore, it is likely that choice of visits is also correlated with ξ . As a result, I use instruments from [Berry et al. \(1995\)](#) (henceforth BLP IVs) and county-dummy-by-fuel-prices (since I have fuel prices at the California level). BLP IVs are standard instruments in the IO literature that use measures of competitiveness in product space to create variation in within-hospice market share and quality choice while being orthogonal to ε_{jt} in the ξ transition equation. In my setting, these are the sum of rivals and sums of rival characteristics. Fuel prices are cost shifters that affect quality choice while being orthogonal to ξ . These instruments are multiplied with the error term ε_{jt} to give moment conditions for GMM. Further details of demand estimation are given in [Appendix E.1](#).

It is now clearer to see why building moments around ε_{jt} is giving me a better exclusion restriction. Without it, I would have to assume that level of competition in the market is orthogonal to the ξ of all incumbents. But that might not be true - if the incumbent has persistently high (or low) ξ , then that could affect entry/exit decisions and hence the level of competition in the market. By allowing for persistence, the claim is that level of competition is orthogonal to some residual term after accounting for persistence.

Table 4 shows the results from the demand estimation. The reputation decay rate τ is estimated to be 0.46, which can be interpreted as saying 54% of a hospice's reputation transfers over to the next year. A positive value of η means that a hospice with higher current and past quality choices is more likely to be chosen by consumers; in other words, a hospice with higher reputation will have higher market share. The unobserved demand shocks are highly persistent (ρ), and consumer preferences within a nest are highly correlated (σ_n). The remaining values in Table 4 gives consumer preferences for hospice

characteristics.²⁰

Figure 1: Impulse response in a duopoly setting to illustrate reputation accumulation.



Notes: Firm 1 and 2 choose identical quality ($a = 0.68$) in every period except periods 3-6, when Firm 1 chooses higher quality ($a = 0.92$). “Baseline share” denotes what Firm 1’s market share would have been had it kept choosing Quality = 0.68 over the same period. Note that there is an outside option so the market shares of the two firms do not sum to 100.

To more easily interpret the value of η and τ , Figure 1 shows an impulse response to convey the importance of reputation effects. The setting is a duopoly. Firm 2 chooses 0.68 every period. Firm 1 chooses 0.68 every period except periods 3-6, when it chooses 0.92. That is, firms 1 and 2 make identical quality choices except in periods 3-6, when firm 1 chooses higher quality.

Figure 1 traces out the market share of firm 1. As firm 1 increases its quality in period 3, its market share rises, but reputation effects mean that its market share keeps rising in the next few periods even as it chooses the same quality level. This shows how reputation accumulates over time. After period 6 it returns to the original quality choice, but its

²⁰I also conduct additional robustness checks for demand estimation. Instead of BLP-IVs, I redo my demand estimation with Gandhi-Houde IVs and obtain similar results. I also use two additional instruments – i) measures of slope cost (i.e. cost of additional visit) for each hospice from the cost data, and ii) Medicare prices. The slope cost acts as a cost shifter. Medicare prices impact quality choice (see counterfactuals) while being orthogonal to firm-specific demand shocks since they are set at the national level and vary by county based on Medicare cost indices. All specifications yield similar results; see Tables 10 and 11 for these and other robustness checks. Finally, a previous version of the paper used “visits-per-patient” as the quality choice variable and obtained similar results.

Table 4: Results of demand estimation.

	Demand
τ	0.463 (0.164)
ρ	0.781 (0.076)
σ_n	0.554 (0.063)
η	0.540 (0.155)
Hospice inpatient unit	0.044 (0.133)
Pediatric program	0.253 (0.078)
Bereavement services	0.012 (0.041)
Day care for adults	0.090 (0.178)
For-profit	-0.232 (0.117)
Agency type: free-standing	-0.204 (0.150)
Agency type: home health based	-0.346 (0.172)
Agency type: hospital-based	-0.290 (0.150)

Notes: The demand model is estimated using 2-step nonlinear GMM with BLP IVs and county-dummy-by-fuel-prices. Robust standard errors are in parentheses. Estimates of county fixed effects are suppressed. See Appendix E.1 for more details.

Also note that the absence of reputation persistence would mean $\tau = 1$; therefore the relevant t-stat for τ is $\frac{\hat{\tau}-1}{s.e.(\tau)}$.

market share remains elevated for several more years and gradually decays down to the original level. This illustrates how reputation decays over time if not maintained with high quality choices.

6.2 Empirical policy function

I regress the quality measure visits-per-patient-day on a variety of covariates to see how quality choice is affected by state variables. This sheds light on how hospices react to reputation levels and product characteristics of themselves and rivals. I interpret this regression as reflecting the policy function used by hospices when making quality decisions.

I use the demand estimates from the previous section to construct reputation stock of each firm in every market-year, the attractiveness of their product attributes ($X'_{jt}\beta_d$), and their unobserved product qualities (ξ_{jt}). In a dynamic game, these values of a hospice and its rivals would enter the policy function.

To account for competitive effects from rivals, I include the following state variables for a given hospice: number of rivals, difference between average reputation of rivals and own, difference between average $X'_{jt}\beta_d$ of rivals and own, and difference between average ξ_{jt} of rivals and own.

When recovering the empirical visit-choice policy function from the data, I try to be cognizant of the issues raised in [Berry and Compiani \(2023\)](#) and [Berry and Compiani \(2021\)](#). Briefly, a regression of visit choice on dynamic states such as reputation stock can suffer from the classic endogeneity problem in the presence of serially correlated unobservables.²¹

To deal with such issues, I take the following steps. First, I include firm-specific fixed effects in my regression. Second, following the suggestions from [Berry and Compiani \(2023\)](#), I use instruments to correct the endogeneity problem. I use lagged $X'_{jt}\beta_d$ and lagged ξ_{jt} to instrument for the dynamic state (difference between average reputation of rivals and own). These instruments are valid because current period's quality choice is affected by *current* $X'_{jt}\beta_d$ and ξ_{jt} , while the last period's reputation stock is affected by last period's $X'_{jt}\beta_d$ and ξ_{jt} . (Note that current period's $X'_{jt}\beta_d$ and ξ_{jt} are included in the

²¹For instance, if the unobserved shocks are serially correlated and also affect the visit choice, then the dynamic state would be influenced by past shocks; past shocks are correlated with current shocks; and so the current period dynamic state is correlated with the current shock.

regression as well.)²²

The resulting 2SLS regression has signs on the state variables that match the general shape of the policy functions I observe in my counterfactuals.²³ Furthermore, I check the 2SLS first-stage regression of dynamic states on the exogenous variables and instruments, and find them to have signs predicted by economic theory; see Appendix C.1 for results and discussion. The results of the regression are given in Table 5. To summarize, hospices choose higher quality when facing more rivals, and when facing rivals with better reputation, $X'_{jt}\beta_d$, and ξ_{jt} . They choose lower quality when they themselves have higher reputation, $X'_{jt}\beta_d$, and ξ_{jt} .

6.3 Cost function from cost data

From the cost data I construct hospice-by-year estimates of per-period fixed cost, intercept cost, and slope cost (adjusted for inflation rates). I regress each on county dummies to obtain county-level averages. Table 6 shows the distribution of regression coefficients, while Table 22 shows the full regression results.

6.4 Estimation of dynamic oligopoly model

As an alternative to using the slope cost estimate (γ_{1j}) from the cost data, I recover both the slope cost estimate and the logit scaling parameter for the hospice's quality choice using the two-stage estimator from [Bajari et al. \(2007\)](#). The first stage involves getting reduced-form estimates of the firms' policy functions from the data, as well as state transition

²²[Berry and Compiani \(2023\)](#) go further and suggest using nonparametric IV methods such as GIV, with their argument being that the policy function of the firm may be highly nonlinear. While I run a linear regression to simplify estimation, I conduct the following check. In my counterfactual, I solve for the policy function for heterogeneous hospices. Then, I check the sign on the policy function for different values of the state variables (own reputation, rival reputation, own and rival $X'_{jt}\beta_d$, own and rival ξ_{jt}). I find that the signs on the solved policy function match the signs on the regression, i.e. they agree on which direction each state variable moves the quality choice. Thus, while the *exact* policy function may be nonlinear, I argue that this regression represents a meaningful approximation of it.

²³In additional counterfactuals (not reported in the main text, available upon request), I solve for the policy function for heterogeneous hospices. I find that, holding other state variables constant, a hospice chooses higher quality if its rivals have high reputation/ $X'_{jt}\beta_d/\xi_{jt}$, and lower quality if the hospice itself has high reputation/ $X'_{jt}\beta_d/\xi_{jt}$.

Table 5: Estimates from 2SLS regression of visit-per-patient-day on covariates.

	Visits-per-patient-day
Count of rivals	0.041 (0.011)
For-profit	0.124 (0.061)
Fuel price	-0.042 (0.022)
Medicare rate	1.92×10^{-5} (7.55×10^{-6})
Average $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$	0.401 (0.151)
Average ξ_{jt} of rivals - own ξ_{jt}	0.455 (0.094)
Average reputation of rivals - own reputation (lagged)	1.589 (0.542)
Firm Fixed Effects	Yes
Obs.	1,608

Notes: Estimates of hospice-specific fixed effects suppressed. The endogenous variable in the regression is the dynamic state “Average reputation of rivals - own reputation (lagged)”. The instruments used are ξ_{jt-1} and $X'_{jt-1}\beta_d$.

Table 6: Distribution of coefficients from regression of cost on county dummies.

	10%	25%	50%	75%	90%
Fixed Cost	213,449.68	360,520.09	534,721.52	919,459.91	1,142,199.24
Intercept	257.40	372.09	587.18	794.80	993.89
Slope	85.82	92.37	109.65	131.13	143.21

Notes: Detailed results are presented in Table 22.

probabilities. In the second stage, these policy functions are used to conduct forward simulation and generate model-predicted conditional choice probabilities (CCPs, explained in Section 4.2.3) for a given guess of cost parameters. Moment conditions are then con-

structured, and an optimizer searches to find cost parameter values that minimize the distance between model-predicted CCPs and observed probabilities. See Appendix E.2 for more details about implementation.

To simplify computation and model solution, I discretize quality choice into 6 tiers. Visits-per-patient-day is discretized into 6 bins between 0.44 to 1.04 visits-per-patient-day (open lower intervals and closed upper intervals) with increments of 0.12. From Table 2 it can be seen that this approximately covers the 10th to 90th percentiles of the observed visits-per-patient-day distribution.

I do not estimate the per-period fixed cost and intercept cost in the dynamic game estimation, instead fixing them at the estimated values from the cost data (see Tables 6 and 22). These are allowed to vary by county; following the notation from earlier, $\phi_j \equiv \phi_{m(j)}$ and $\gamma_j \equiv \gamma_{0m(j)}$, where $m(j)$ is the county where hospice j is located. Per-period fixed cost is not identified since I do not model exit, and in simulations I found that the cost function intercept was not clearly estimated either.

The first stage of Bajari et al. (2007) requires an estimate of the policy function from the data. To this end, I use a variant of the empirical policy function from Section 6.2. I make the following adjustments from the empirical policy function. First, recall that I discretize visit choice into 6 tiers; as a result, I use the empirical policy function and random draws from the residual distribution to make predictions of quality choice given values of state variables, then round it to the nearest quality tier. Second, I do not make use of the firm-specific fixed effects for making predictions. Most firms are observed for a few periods, and so the fixed effect estimates are likely to be biased. Also, as alluded to earlier, the fixed effects are identified off of an assumption of linear policy function, which is unlikely to hold in practice.²⁴

The state variables that enter a hospice's value function are its hospice characteristics and unobserved shocks, its rivals' characteristics and unobserved shocks, its lagged reputation, its rivals' lagged reputations, Medicare reimbursement rates, fixed costs, intercept costs, market size, and identity of the county where it operates.

²⁴Alternatively, I can use a flexible function of the same state variables to approximate the policy function. The advantage is this better captures any nonlinearities that the policy function may exhibit; the disadvantage is I cannot control for serially correlated unobservables.

Next I explain how I implement state transitions in my forward simulation. For the estimation, I assume that a firm’s characteristics do not change over time for simplification; this assumption can be relaxed. Firms’ current period unobserved shocks (ξ) are imputed from the demand estimation; the estimated persistence rate is used to simulate it forward while drawing the errors (ε) from the econometric error distribution. Market size and Medicare rates follow an AR(1) process that is estimated from the data. From my data I find the average length-of-stay $\overline{LOS}$ to be 49.16 days, and I set the discount rate $\beta = 0.95$.

Entry and exit are infrequent in my data, so I implement entry and exit as exogenous transitions rather than endogenizing them as part of the firm’s choice problem. In my forward simulation, entry by potential entrants into a market is modeled through an ordered logit specified in Table 7. Exit by a firm happens with a probability of 4% every period, which matches exit rates in my data.

Table 7: Ordered logit of entry count on market characteristics, used as entry transition function.

	Entry count
Firm count	-0.300 (0.117)
Market size	0.001 (3.461×10^{-4})
Medicare price	-3.846×10^{-4} (2.423×10^{-4})
County FE	Yes

Notes: For each market-year, I calculate the total number of entries (“entry count”). Then I run an ordered logit of entry counts on market characteristics (market fixed effects, market size, Medicare price). I restrict the ordered logit to market-years with 3 or less entries; greater than 3 entries are very rare and give unintuitive results. Estimates of county fixed effects and cutoffs suppressed.

The first-stage policy functions and transition matrices are used in forward simulation to construct choice-specific value functions. This combines methods discussed in [Bajari et al. \(2007\)](#) and [Pakes et al. \(2007\)](#). See Appendix E.2 for a description of the estimation procedure and Appendix D for an explanation of how I impute reputation stocks of older hospices.

The simulated path continues for 80 periods. For each action and each observed state I run the forward simulation 200 times. The resulting choice-specific value functions are then used to construct the model-predicted choice probabilities.

Next, I describe how these model-predicted choice probabilities are used to estimate the hospice cost function. The intuition is that I search for values of parameters θ that minimize the distance between model-predicted choice probabilities and observed choices.

The dynamic oligopoly model is estimated using two-step GMM. The GMM residual term for observation n is:

$$\Xi_n(\theta) = a_n^{data} - \sum_{a_n \in \mathcal{A}} a_n \hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$$

where $\hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$ is the predicted choice probability for action a_n , a_n^{data} is the observed quality choice for observation n , θ is the guess of structural parameters, and $\mathbf{x}_n^{data}$ is the market state for observation n . The error term measures how close the average model-predicted choice is to the observed choice. The instruments used in GMM are the county indicators and the variables in the first-stage empirical policy functions (with the exception of the firm fixed effects), giving us the moment conditions $E[Z_n' \Xi_n(\theta)]$. The intuition for this is that the model's predicted choices leverage these variables, and so at the true parameter values the model's prediction errors should be orthogonal to these variables. The resulting GMM optimization problem is:

$$\min_{\theta} \left[\frac{1}{N} \sum_n Z_n' \Xi_n(\theta) \right]' \hat{W} \left[\frac{1}{N} \sum_n Z_n' \Xi_n(\theta) \right]$$

This objective function is minimized to estimate cost parameters of hospices (γ_1 and θ_e). Standard errors are calculated via block-bootstrapping, similar to [Wang \(2022\)](#). I sample blocks of county-year observations from my data with replacement until I match the total number of county-year observations in my dataset. Then I re-estimate the model on each bootstrap sample. I use the sample variance over all the resulting bootstrap estimates to calculate the standard errors for my cost parameters. 400 bootstrap runs are used.

Table 8 gives the results of the dynamic supply estimation. I find that the cost of an additional visit is approximately \$192 (recall that per-period fixed cost and the intercept

term are obtained from the cost data and used in the [Bajari et al. \(2007\)](#) algorithm.) This is higher than my measure of visit cost from the expense data (see Table 6), which had a median of \$109.65. The higher value from my [Bajari et al. \(2007\)](#) estimation makes sense because it measure the economic cost, while the cost measures from expense data reflect accounting cost. Economic cost should be higher as it includes, in addition to accounting cost, factors such as opportunity cost and presence of constraints. In my policy counterfactuals I will therefore use the cost measure from the [Bajari et al. \(2007\)](#) estimation.²⁵

Table 8: Results of supply estimation.

γ_1	192.252 (46.110)
θ_e	1.066×10^6 (7.185×10^6)

Notes: The slope cost γ_1 and logit scaling parameter θ_e are estimated using the method of [Bajari et al. \(2007\)](#). Standard errors (in parentheses) are calculated via bootstrapping at the county-year level. 400 bootstrap runs were used.

7 Policy counterfactuals

Given the above estimates of the structural parameters, I can now conduct counterfactuals to investigate the effects of various policy tools. I plot the policy function against various state variables; among others, I find evidence of reputation building and milking incentives. As reputation becomes more persistent, hospices choose higher quality. Lower Medicare reimbursement leads to worsening hospice quality. Compared to Medicare's

²⁵I interpret the cost estimate from the dynamic game estimation as averaging the economic cost of an additional visit over counties and hospices, as the accounting estimates suggest there is probable heterogeneity at least across counties. I could alternatively run my counterfactuals using the accounting cost estimates, and preliminary runs suggest very similar qualitative results.

I also interpret my cost estimates as subsuming any altruism by the hospice. That is, greater altruism by a hospice will lead to lower cost estimates – the hospices will be choosing higher quality than would be suggested by their accounting costs, and my estimation will rationalize this as the hospice having lower cost. See [Hackmann \(2019\)](#) for a paper that backs out the altruism parameter from the preference function.

While Table 8 shows that θ_e is noisily estimated, I find that the objective function is convex in γ_1 and θ_e .

current per-day reimbursement scheme, a hybrid per-day per-visit hospice reimbursement scheme achieves the same equilibrium quality with lower Medicare spending.

7.1 Solving for equilibrium outcomes

To conduct counterfactuals I need to solve the model and recover the policy function of the hospice. This requires me to calculate $\Psi(a_j, \mathbf{x}, \sigma^*)$, $\hat{v}(a_j, \mathbf{x}, \sigma^*)$ and $V(\mathbf{x}, \sigma^*)$ such that Equations (9), (10), and (11) simultaneously hold at every point in the state space. I solve the model using the policy iteration method (Rust (1994), Judd (1998), Benkard (2004), Caoui and Steck (2023)). This gives me the conditional choice probabilities and ex-ante value function at every action-state combination. For more details on model solution, see Appendix E.3.

To ease computational burden, I solve a simpler version of the model. First, I limit the number of firms to 2 or 3. This is a good approximation of the competitive environment in my data, since firm-count at the market-year level has a median of 2. Second, I do not allow for entry. This rules out firms acquiring reputation for entry deterrence. This is a reasonable assumption to make because I find no evidence of entry deterrence through reputation in my data, as well as because entry happens in less than 10% of market-year observations. Third, I do not allow for exit, again because of how infrequently it is observed in the data.

While not explicitly modeling entry or exit, I report the ex-ante value functions of incumbents for all policy simulations. Changes in ex-ante value functions can be indicative of entry-exit incentives; for instance, a policy change that lowers ex-ante value functions may induce exit, while an increase may incentivize entry. In Appendix F.1 I discuss the intuition behind this approach as well as what it rules out.

Another decision to simplify computation is to discretize the reputation stock, and impose a stochastic reputation evolution process to minimize the effects of any such discretization. This closely follows Benkard (2004). I discretize reputation into bins from 0.0 to 1.5 with increments of 0.3; this range matches the reputation stocks observed in my dataset, and finer discretizations have yielded the same results.

The stochastic reputation evolution works as follows. Fix a firm, an industry state, and

a chosen action a_t . Let ψ_t be the reputation stock predicted by the reputation transition equation $\psi_t = (1 - \tau)\psi_{t-1} + \eta a_t$. Let ψ_d be the largest reputation stock less than ψ_t and ψ_u be the lowest reputation stock greater than ψ_t . In the counterfactual, the attained reputation stock ψ_t^* then follows the stochastic process below:

$$\psi_t^* = \begin{cases} \psi_u & \text{w.p. } \frac{\psi_t - \psi_d}{\psi_u - \psi_d} \\ \psi_d & \text{w.p. } 1 - \frac{\psi_t - \psi_d}{\psi_u - \psi_d} \end{cases} \quad (13)$$

Counterfactual results are reported as follows. For a given policy experiment, I solve the model to obtain the CCPs. The CCP gives the probability of a firm making each action choice for any given state under the chosen policy experiment. I pick a starting point for a market, then simulate forward 1000 periods using the policy function. I report the average quality choice by one firm over the simulation. To prevent my choice of initial conditions from contaminating the result, the first 200 periods are dropped; additionally I check by visual inspection to ensure that the industry has settled into its long-run state.²⁶

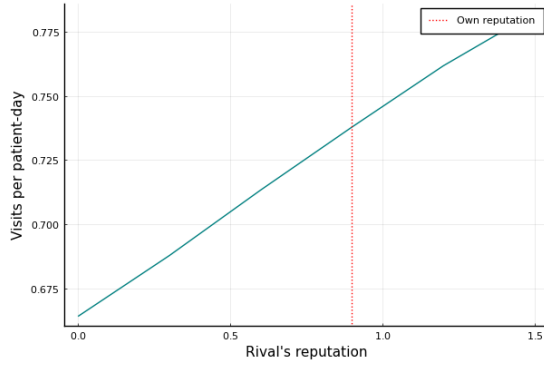
7.2 Illustrating the policy function

To illustrate the shape of the policy function, I solve the model for 2 identical hospices and plot the policy function. The y-axis reports the CCP-weighted quality choice (i.e. $\sum_{a_j \in \mathcal{A}} \Psi(a_j, \cdot, \sigma^*) a_j$) for the corresponding state in the x-axis.

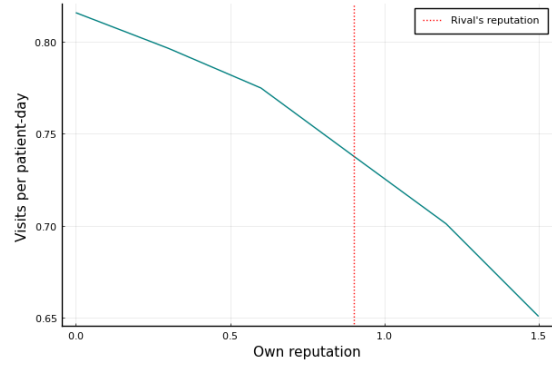
1. Figure 2a holds reputation of a hospice fixed, and shows that a hospice raises its quality as it faces a rival with higher and higher reputation.
2. Figure 2b holds reputation of the rival hospice fixed, and shows that a hospice lowers quality as its own reputation improves. This suggests reputation “building and milking” incentives that have been explored in the theoretical literature - once a hospice builds up high reputation compared to its rivals, it chooses low quality for a while; as its reputation drops, it again chooses high quality to build it up.

²⁶While I show how different policy levers can affect equilibrium quality, I do not specify what the quality target should be. This is because my demand model is silent about consumer valuation of visits. The thought experiment is that CMS has figured out the quality target through studies of their own, and can achieve said target with the policy instruments I explore.

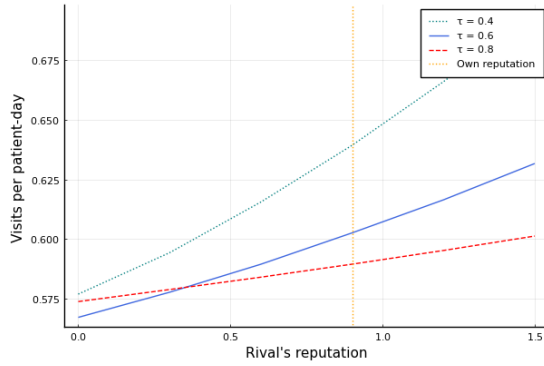
Figure 2: Plotting the policy function against state variables and different τ values.



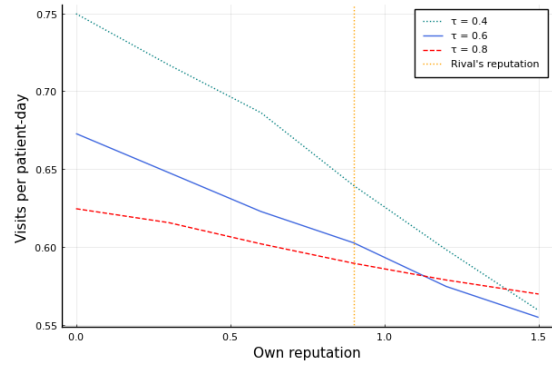
(a) Policy function against rival reputation



(b) Policy function against own reputation



(c) Policy function against rival reputation for different τ

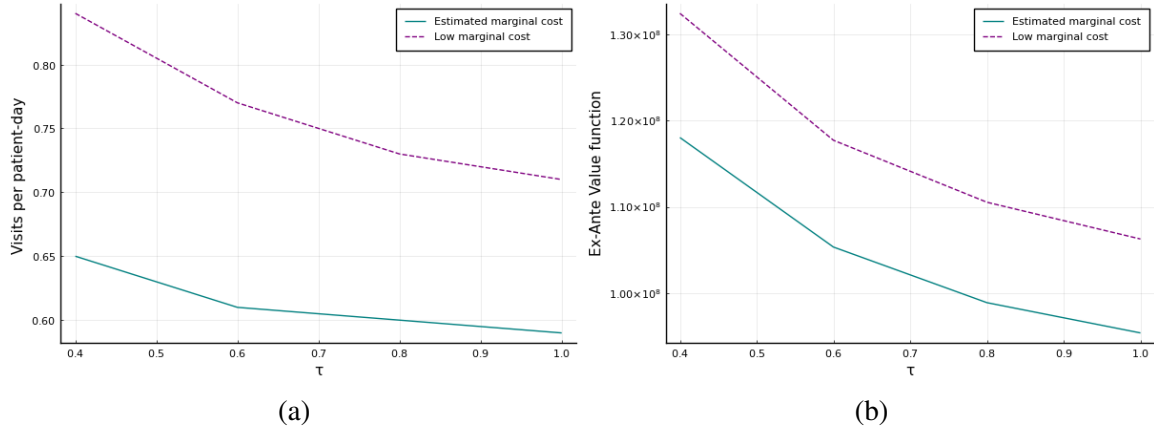


(d) Policy function against own reputation for different τ

Notes: The model is solved for two identical hospices via policy iteration. The y-axis reports $\sum_{a_j \in \mathcal{A}} \Psi(a_j, x_m, \sigma^*) \cdot a_j$, which gives the CCP-weighted quality choices. Figures 2a and 2c hold the hospice's own reputation fixed at 0.9. Figures 2b and 2d hold the rival hospice's reputation fixed at 0.9.

- Figures 2c and 2d show that as reputation depreciates faster (τ gets larger) the policy functions become flatter with respect to own reputation and rival reputation.

Figure 3: Quality choice in a duopoly against τ for different marginal costs.



Notes: The figure reports the average quality choice and average ex-ante value function of one hospice over the simulation. The model is solved for two identical hospices via policy iteration.

7.3 Increasing persistence of reputation

The first set of counterfactuals involve the persistence of reputation. From the perspective of policymaking, this can tell us whether policy changes that lead to greater reputation persistence can improve consumer welfare. For instance, in 2017 CMS launched Hospice Compare, a website that details quality information about hospices. Such review websites can aggregate and sustain information on hospices' quality choices for much longer and with greater accuracy, and can be thought of as helping reputation persist longer. However, the website has seen limited consumer participation and awareness. Figure 3a looks at how the visits-per-patient-day varies as reputation becomes less persistent. The counterfactual is done for two identical firms competing in a market.

The takeaway is that hospice quality falls as reputation decays faster. The intuition is that the marginal gain from increasing quality is smaller: since reputation is decaying more quickly, an increase in quality leads to a smaller increase in future sales, so it is no longer optimal to incur the higher cost from higher quality.

Given that we saw heterogeneity across markets when it came to slope cost (Table 6), I also study how this graph changes if the hospices were to have lower marginal cost. Hospices choose higher equilibrium quality when costs are lower; however, perhaps sur-

prisingly, I see that hospices react more strongly to a change in the reputation depreciation rate. That is, for lower marginal cost, equilibrium quality *drops* by more as τ goes from 0.4 to 1.0. This suggests that improving reputation persistence will have heterogeneous effects across markets based on the underlying cost primitives, with lower marginal cost regions seeing stronger quality responses.

7.4 Increasing Medicare reimbursement rates

To see how hospices react to higher/lower reimbursement rates, my second set of counterfactuals study how hospice quality changes as Medicare rate increases. This is motivated by the fact that hospices (and other healthcare providers) have frequently complained Medicare reimbursement rates are too low and have not kept up with inflation rates.

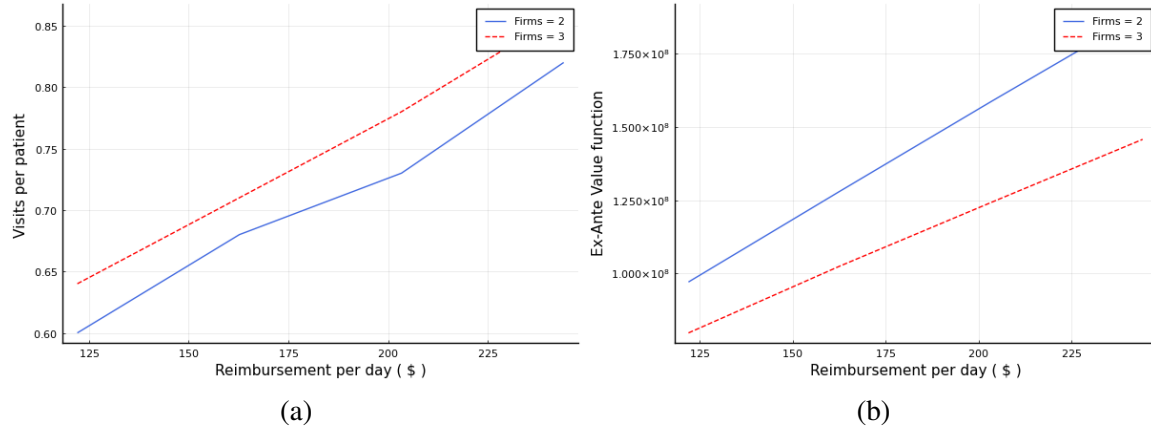
Figure 4a reports a counterfactual where 2 and 3 identical firms make quality choices under different Medicare reimbursement rates. It can be seen that i) higher Medicare prices lead to higher qualities being chosen by each firm, and ii) quality choice is higher with more firms due to greater competition.

Lower Medicare reimbursement rates can also influence equilibrium quality choice by affecting entry and exit incentives. Figure 4b shows how the average ex-ante value function of a hospice over the simulation changes as Medicare rate increases. Higher (lower) Medicare rate leads to higher (lower) ex-ante value function, which may lead to entry (exit). The change in the number of firms therefore leads to an additional impact on quality. In other words, higher Medicare rates can lead to higher quality both through more intense competition between existing firms as well as through entry of new firms.

7.5 Tying Medicare reimbursement to quality choice

My final set of counterfactuals investigates alternative reimbursement schemes for Medicare. The current reimbursement scheme involves Medicare paying hospices for every day a patient is enrolled, i.e. a per-day per-patient scheme. If we want to achieve a target quality level at the lowest possible Medicare spending, a natural route to investigate is to tie part of the reimbursement to quality.

Figure 4: Quality choice against different Medicare reimbursement rates.



Notes: The figure reports the average quality choice and average ex-ante value function of one hospice over the simulation. The model is solved for two and three identical hospices via policy iteration.

To that end, I solve the model for hybrid per-day per-visit schemes, with the results shown in Table 9. I progressively reduce the per-day rate and increase the per-visit rate while ensuring that under the new terms, the equilibrium quality choice remains at 0.78 visits-per-patient-day. The total Medicare spending is calculated for each scheme; for clarity, I normalize the spending at each scheme by the total Medicare spending at the per-day per-patient rate (i.e. Medicare's current scheme).

I find that Medicare can reduce per-day rate and raise per-visit rate in such a way that total spending declines even as equilibrium quality choice stays the same. The takeaway is that potential cost savings can be achieved by shifting weight from per-day to per-visit reimbursement. Note that how far I can increase the per-visit rate is limited by the estimated marginal cost, since a per-visit rate exceeding the marginal cost will lead to

Table 9: Reimbursement schemes that lead to 0.78 visits-per-patient-day in a duopoly.

Per-day rate	Per-visit rate	Medicare Spending (normalized)	Ex-Ante Value Function (normalized)
223.7	0.0	1.0	1.0
203.4	20.0	0.97	0.93
183.1	40.0	0.96	0.86
142.4	80.0	0.91	0.71
101.7	120.0	0.87	0.57
61.0	160.0	0.83	0.43

Notes: The rates are in US dollars. The total Medicare spending associated with each scheme is normalized with respect to the spending of the per-day scheme. The average ex-ante value function is normalized with respect to the ex-ante value function of the per-day scheme. As we move down rows, equilibrium quality stays the same yet Medicare spending declines.

hospices making unlimited visits to maximize profits.²⁷

There is a trade-off associated with this scheme however - as we change the reimbursement structure in this direction, the average ex-ante value function of each hospice decreases, indicating that hospices are making lower profits. As a result, this scheme may risk inducing exit among hospices. Therefore, when implementing this scheme, the policymakers have to be wary of not inducing exit and worsening equilibrium quality.²⁸

²⁷A possible issue with subsidizing visits is that it could encourage spurious visits by hospices. While my model cannot allow for firms making spurious visits, it is interesting to think about how this might play out in the real world. First, since reputation effects are salient in this industry, a hospice that makes spurious visits can develop a bad reputation and lose future consumers, so reputation effects might rein in such issues. Second, Medicare can take additional steps to discourage spurious visits. It can adopt finer contract structures (such as reimbursing hospices for up to 3 visits a week). It can also enforce stronger monitoring. For instance, the patient's family may be asked to sign off after every visit. Medicare can also deploy the sort of documentation and monitoring that it already employs in other settings such as DRGs.

²⁸I repeat this counterfactual with 3 identical hospices and obtain the same qualitative findings; see Appendix C.2.

8 Conclusion

Using firm-level data from California, I study quality choice by hospices, uncover the importance of hospice reputation for consumers, and explore counterfactual policies that can incentivize higher hospice quality and lower Medicare spending. I build and estimate a structural model of consumer demand and hospice quality choice to do so. My measure of hospice quality is the average number of visits made by a hospice to its patients per day in a year, and I define reputation of a hospice to be a function of its current and past quality choices. As a result, a hospice can accumulate reputation over time by consistently choosing high quality. I use my structural model to quantify the importance of reputation for consumers choosing hospices and estimate the hospice cost function. Reputation plays an important role in consumer demand for hospices, and past reputation decays at an annual rate of 46%. The cost of an additional visit by a hospice to a patient is approximately \$192. Counterfactuals show that hospice quality increases with higher Medicare prices and greater persistence of reputation. I compare the current per-day Medicare reimbursement scheme with a hybrid per-day per-visit scheme, and find that the latter can incentivize the same level of hospice quality at lower Medicare spending.

References

- Aguirregabiria, Victor, Allan Collard-Wexler, and Stephen P. Ryan (2021) “Dynamic games in empirical industrial organization,” *Handbook of Industrial Organization*, 4, 225–343, DOI: [10.1016/BS.HESIND.2021.11.004](https://doi.org/10.1016/BS.HESIND.2021.11.004).
- Aguirregabiria, Victor and Pedro Mira (2007) “Sequential estimation of dynamic discrete games,” *Econometrica*, 75, 1–53, DOI: [10.1111/J.1468-0262.2007.00731.X](https://doi.org/10.1111/J.1468-0262.2007.00731.X).
- Alam, Rubaiyat and Rena Conti (2024) “Entry and Pricing with Fighting Brands: Evidence from Generics and Authorized Generics,” *Working paper*.
- Arcidiacono, Peter and Paul B. Ellickson (2011) “Practical methods for estimation of dynamic discrete choice models,” *Annual Review of Economics*, 3, 363–394, DOI: [10.1146/ANNUREV-ECONOMICS-111809-125038/CITE/REFWORKS](https://doi.org/10.1146/ANNUREV-ECONOMICS-111809-125038/CITE/REFWORKS).

- Bachmann, Rüdiger, Gabriel Ehrlich, Ying Fan, Dimitrije Ruzic, and Benjamin Leard (2023) “Firms and Collective Reputation: a Study of the Volkswagen Emissions Scandal,” *Journal of the European Economic Association*, 21, 484–525, DOI: [10.1093/JEEA/JVAC046](https://doi.org/10.1093/JEEA/JVAC046).
- Bai, Jie (2024) “Melons as Lemons: Asymmetric Information, Consumer Learning and Seller Reputation,” *Working paper*.
- Bajari, Patrick, C. Lanier Benkard, and Jonathan Levin (2007) “Estimating Dynamic Models of Imperfect Competition,” *Econometrica*, 75, 1331–1370, DOI: [10.1111/J.1468-0262.2007.00796.X](https://doi.org/10.1111/J.1468-0262.2007.00796.X).
- Bar-Isaac, Heski and Steven Tadelis (2008) “Seller reputation,” *Foundations and Trends in Microeconomics*, 4, 273–351, DOI: [10.1561/07000000027](https://doi.org/10.1561/07000000027).
- Benkard, C. Lanier (2004) “A Dynamic Analysis of the Market for Wide-Bodied Commercial Aircraft,” *The Review of Economic Studies*, 71, 581–611, DOI: [10.1111/J.1467-937X.2004.00297.X](https://doi.org/10.1111/J.1467-937X.2004.00297.X).
- Berry, Steven, James Levinsohn, and Ariel Pakes (1995) “Automobile Prices in Market Equilibrium,” *Econometrica*, 63, 841, DOI: [10.2307/2171802](https://doi.org/10.2307/2171802).
- Berry, Steven T. (1994) “Estimating Discrete-Choice Models of Product Differentiation,” *The RAND Journal of Economics*, 25, 242, DOI: [10.2307/2555829](https://doi.org/10.2307/2555829).
- Berry, Steven T. and Giovanni Compiani (2021) “Empirical Models of Industry Dynamics with Endogenous Market Structure,” *Annual Review of Economics*, 13, 309–334, DOI: [10.1146/ANNUREV-ECONOMICS-081720-120019/CITE/REFWORKS](https://doi.org/10.1146/ANNUREV-ECONOMICS-081720-120019/CITE/REFWORKS).
- Berry, Steven T. and Giovanni Compiani (2023) “An Instrumental Variable Approach to Dynamic Models,” *The Review of Economic Studies*, 90, 1724–1758, DOI: [10.1093/RESTUD/RDAC061](https://doi.org/10.1093/RESTUD/RDAC061).
- Borkovsky, Ron N., Avi Goldfarb, Avery M. Haviv, and Sridhar Moorthy (2017) “Measuring and Understanding Brand Value in a Dynamic Model of Brand Management,”

- <https://doi.org/10.1287/mksc.2016.1020>, 36, 471–499, DOI: [10.1287/MKSC.2016.1020](https://doi.org/10.1287/MKSC.2016.1020).
- Bronnenberg, Bart J., Jean-Pierre Dubé, and Chad Syverson (2022) “Marketing Investment and Intangible Brand Capital,” *Journal of Economic Perspectives*, 36, 53–74, DOI: [10.1257/JEP.36.3.53](https://doi.org/10.1257/JEP.36.3.53).
- Cabral, Luís and Ali Hortaçsu (2010) “The Dynamics of Seller Reputation: Evidence from eBay,” *The Journal of Industrial Economics*, 58, 54–78, DOI: [10.1111/J.1467-6451.2010.00405.X](https://doi.org/10.1111/J.1467-6451.2010.00405.X).
- Camarda, Enrico (2022) “Welfare effects of outcome-based prices in the English National Health Service,” *Working paper*.
- Caoui, El Hadi and Andrew Steck (2023) “Diversification, Market Entry, and the Global Internet Backbone,” *SSRN Electronic Journal*, DOI: [10.2139/SSRN.4478868](https://doi.org/10.2139/SSRN.4478868).
- Chung, Andrea P and Alan Sorensen (2018) “For-Profit Entry and Market Expansion in the Hospice Industry.”
- Collard-Wexler, Allan (2013) “Demand Fluctuations in the Ready-Mix Concrete Industry,” *Econometrica*, 81, 1003–1037, DOI: [10.3982/ECTA6877](https://doi.org/10.3982/ECTA6877).
- Dalton, Christina Marsh and W. David Bradford (2019) “Better together: Coexistence of for-profit and nonprofit firms with an application to the U.S. hospice industry,” *Journal of Health Economics*, 63, 1–18, DOI: [10.1016/J.JHEALECO.2018.10.001](https://doi.org/10.1016/J.JHEALECO.2018.10.001).
- Doraszelski, Ulrich and Ariel Pakes (2007) “Chapter 30 A Framework for Applied Dynamic Analysis in IO,” *Handbook of Industrial Organization*, 3, 1887–1966, DOI: [10.1016/S1573-448X\(06\)03030-5](https://doi.org/10.1016/S1573-448X(06)03030-5).
- Dubé, Jean Pierre, Günter J. Hitsch, and Puneet Manchanda (2005) “An Empirical Model of Advertising Dynamics,” *Quantitative Marketing and Economics* 2005 3:2, 3, 107–144, DOI: [10.1007/S11129-005-0334-2](https://doi.org/10.1007/S11129-005-0334-2).

- Einav, Liran, Amy Finkelstein, and Neale Mahoney (2018) “Provider Incentives and Healthcare Costs: Evidence From Long-Term Care Hospitals,” *Econometrica*, 86, 2161–2219, DOI: [10.3982/ECTA15022](https://doi.org/10.3982/ECTA15022).
- Ericson, Richard and Ariel Pakes (1995) “Markov-Perfect Industry Dynamics: A Framework for Empirical Work,” *The Review of Economic Studies*, 62, 53–82, DOI: [10.2307/2297841](https://doi.org/10.2307/2297841).
- Fan, Ying, Jiandong Ju, and Mo Xiao (2016) “Reputation premium and reputation management: Evidence from the largest e-commerce platform in China,” *International Journal of Industrial Organization*, 46, 63–76, DOI: [10.1016/J.IJINDORG.2016.01.004](https://doi.org/10.1016/J.IJINDORG.2016.01.004).
- Gandhi, Amit and Jean-Francois Houde (2019) “Measuring Substitution Patterns in Differentiated-Products Industries.”
- Gaynor, Martin (2006) “What Do We Know About Competition and Quality in Health Care Markets?,” DOI: [10.3386/W12301](https://doi.org/10.3386/W12301).
- Gaynor, Martin, Rodrigo Moreno-Serra, and Carol Propper (2013) “Death by Market Power: Reform, Competition, and Patient Outcomes in the National Health Service,” *American Economic Journal: Economic Policy*, 5, 134–66, DOI: [10.1257/POL.5.4.134](https://doi.org/10.1257/POL.5.4.134).
- Grennan, Matthew (2013) “Price Discrimination and Bargaining: Empirical Evidence from Medical Devices,” *American Economic Review*, 103, 145–77, DOI: [10.1257/AER.103.1.145](https://doi.org/10.1257/AER.103.1.145).
- Grieco, Paul L.E. and Ryan C. Mcdevitt (2017) “Productivity and Quality in Health Care: Evidence from the Dialysis Industry,” *The Review of Economic Studies*, 84, 1071–1105, DOI: [10.1093/RESTUD/RDW042](https://doi.org/10.1093/RESTUD/RDW042).
- Gruber, Jonathan, David H Howard, Jetson Leder-Luis Theodore, L Caputi, Jetson Leder-Luis, and Theodore L Caputi (2023) “Dying or Lying? For-Profit Hospices and End of Life Care,” DOI: [10.3386/W31035](https://doi.org/10.3386/W31035).

- Hackmann, Martin B. (2019) “Incentivizing Better Quality of Care: The Role of Medicaid and Competition in the Nursing Home Industry,” *American Economic Review*, 109, 1684–1716, DOI: [10.1257/AER.20151057](https://doi.org/10.1257/AER.20151057).
- Hotz, V. Joseph and Mo Xiao (2011) “The Impact of Regulations on the Supply and Quality of Care in Child Care Markets,” *American Economic Review*, 101, 1775–1805, DOI: [10.1257/AER.101.5.1775](https://doi.org/10.1257/AER.101.5.1775).
- Igami, Mitsuru (2018) “Industry Dynamics of Offshoring: The Case of Hard Disk Drives,” *American Economic Journal: Microeconomics*, 10, 67–101, DOI: [10.1257/MIC.20150256](https://doi.org/10.1257/MIC.20150256).
- Jin, Ginger Zhe and Phillip Leslie (2009) “Reputational Incentives for Restaurant Hygiene,” *American Economic Journal: Microeconomics*, 1, 237–67, DOI: [10.1257/MIC.1.1.237](https://doi.org/10.1257/MIC.1.1.237).
- Judd, Kenneth L. (1998) “Numerical Methods in Economics,” 656.
- Lee, Robin S. (2013) “Vertical Integration and Exclusivity in Platform and Two-Sided Markets,” *American Economic Review*, 103, 2960–3000, DOI: [10.1257/AER.103.7.2960](https://doi.org/10.1257/AER.103.7.2960).
- Lin, Haizhen (2015) “Quality Choice and Market Structure: A Dynamic Analysis of Nursing Home Oligopolies,” *International Economic Review*, 56, 1261–1290, DOI: [10.1111/IERE.12137](https://doi.org/10.1111/IERE.12137).
- Newberry, Peter and Xiaolu Zhou (2019) “Heterogeneous Effects of Online Reputation for Local and National Retailers,” *International Economic Review*, 60, 1565–1587, DOI: [10.1111/IERE.12397](https://doi.org/10.1111/IERE.12397).
- Olley, G. Steven and Ariel Pakes (1996) “The Dynamics of Productivity in the Telecommunications Equipment Industry,” *Econometrica*, 64, 1263, DOI: [10.2307/2171831](https://doi.org/10.2307/2171831).
- Pakes, Ariel, Michael Ostrovsky, and Steven Berry (2007) “Simple estimators for the parameters of discrete dynamic games (with entry/exit examples),” *The RAND Journal of Economics*, 38, 373–399, DOI: [10.1111/J.1756-2171.2007.TB00073.X](https://doi.org/10.1111/J.1756-2171.2007.TB00073.X).

- Rust, John (1987) “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher,” *Econometrica*, 55, 999, DOI: [10.2307/1911259](https://doi.org/10.2307/1911259).
- Rust, John (1994) “Chapter 51 Structural estimation of markov decision processes,” *Handbook of Econometrics*, 4, 3081–3143, DOI: [10.1016/S1573-4412\(05\)80020-0](https://doi.org/10.1016/S1573-4412(05)80020-0).
- Ryan, Stephen P. (2012) “The Costs of Environmental Regulation in a Concentrated Industry,” *Econometrica*, 80, 1019–1061, DOI: [10.3982/ECTA6750](https://doi.org/10.3982/ECTA6750).
- Saeedi, Maryam (2019) “Reputation and adverse selection: theory and evidence from eBay,” *The RAND Journal of Economics*, 50, 822–853, DOI: [10.1111/1756-2171.12297](https://doi.org/10.1111/1756-2171.12297).
- Tadelis, Steven (2016) “Reputation and Feedback Systems in Online Platform Markets,” <http://dx.doi.org/10.1146/annurev-economics-080315-015325>, 8, 321–340, DOI: [10.1146/ANNUREV-ECONOMICS-080315-015325](https://doi.org/10.1146/ANNUREV-ECONOMICS-080315-015325).
- Wang, Yanfei (2022) “Competition and Multilevel Technology Adoption: A Dynamic Analysis of Electronic Medical Records Adoption in U.S. Hospitals,” *International Economic Review*, 63, 1357–1395, DOI: [10.1111/IERE.12586](https://doi.org/10.1111/IERE.12586).
- Xiao, Mo (2010) “Is quality accreditation effective? Evidence from the childcare market,” *International Journal of Industrial Organization*, 28, 708–721, DOI: [10.1016/J.IJINDORG.2010.04.004](https://doi.org/10.1016/J.IJINDORG.2010.04.004).

Supplementary Appendices for Online Publication

A Robustness checks for demand estimation

I estimate the following system of equations using 2-step GMM.

$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha_{n(j)} + X'_{jt}\beta_d + \sigma_n \ln(s_{j|gt}) + \xi_{jt} + \eta[a_{jt} + (1 - \tau)a_{jt-1} + (1 - \tau)^2 a_{jt-2} + \dots]$$

$$\xi_{jt} = \rho \xi_{jt-1} + \varepsilon_{jt}$$

Here τ denotes depreciation rate of reputation, ρ denotes persistence of unobserved hospice-specific demand shocks, σ_n is the nest parameter, and η is the impact of quality choice on the reputation stock.

For robustness, I re-estimate this model using different sets of instruments. I continue to find evidence of reputation and its persistence.

Also note that the absence of reputation persistence would mean $\tau = 1$; therefore the t-stat for τ is $\frac{\hat{\tau}-1}{s.e.(\hat{\tau})}$.

Table 10: Results of demand estimation with BLP-IVs.

	(1)	(2)	(3)
τ	0.345 (0.145)	0.460 (0.143)	0.142 (0.111)
ρ	0.767 (0.066)	0.478 (0.150)	0.444 (0.164)
σ	0.601 (0.056)	0.584 (0.063)	0.685 (0.078)
η	0.400 (0.106)	0.626 (0.191)	0.230 (0.117)
Hospice inpatient unit	0.071 (0.168)	0.113 (0.080)	0.142 (0.081)
Pediatric program	0.241 (0.084)	0.282 (0.086)	0.212 (0.076)
Bereavement services	-0.013 (0.045)	0.010 (0.038)	-0.025 (0.033)
Day care for adults	0.032 (0.159)	0.112 (0.205)	-0.066 (0.123)
For-profit	-0.164 (0.102)	-0.216 (0.083)	-0.035 (0.114)
Agency type: free-standing	-0.222 (0.152)	-0.191 (0.114)	-0.176 (0.091)
Agency type: home health based	-0.316 (0.201)	-0.238 (0.150)	-0.166 (0.139)
Agency type: hospital-based	-0.277 (0.163)	-0.205 (0.131)	-0.182 (0.121)

Notes: (1): BLP-IV with county-by-fuel IVs and log-slope-per-visit and inflation-adjusted Medicare prices; (2): BLP-IV with log-slope-per-visit and inflation-adjusted Medicare prices; (3): BLP-IV with log-slope-per-visit. The demand model is estimated using 2-step nonlinear GMM. Robust standard errors are in parentheses. Estimates of county fixed effects are suppressed.

Table 11: Results of demand estimation with GHIVs.

	(1)	(2)	(3)	(4)
τ	0.270 (0.136)	0.097 (0.062)	0.256 (0.136)	0.329 (0.160)
ρ	0.886 (0.098)	0.861 (0.063)	0.893 (0.070)	0.629 (0.152)
σ	0.578 (0.085)	0.685 (0.082)	0.630 (0.075)	0.581 (0.116)
η	0.319 (0.149)	0.164 (0.074)	0.274 (0.080)	0.414 (0.206)
Hospice inpatient unit	0.102 (0.185)	0.094 (0.158)	0.094 (0.173)	0.124 (0.103)
Pediatric program	0.238 (0.110)	0.146 (0.099)	0.227 (0.098)	0.296 (0.091)
Bereavement services	0.003 (0.056)	-0.034 (0.042)	-0.009 (0.051)	-0.008 (0.047)
Day care for adults	-0.043 (0.218)	-0.083 (0.108)	-0.032 (0.115)	-0.078 (0.299)
For-profit	-0.179 (0.239)	-0.027 (0.168)	-0.134 (0.203)	-0.200 (0.120)
Agency type: free-standing	-0.217 (0.273)	-0.200 (0.162)	-0.205 (0.222)	-0.169 (0.115)
Agency type: home health based	-0.358 (0.277)	-0.255 (0.186)	-0.306 (0.250)	-0.220 (0.166)
Agency type: hospital-based	-0.318 (0.260)	-0.244 (0.178)	-0.275 (0.257)	-0.222 (0.143)

Notes: In what follows, GHIV stands for Gandhi-Houde IVs ([Gandhi and Houde \(2019\)](#)). (1): GHIV with county-fuel IVs (2): GHIV with county-fuel IVs and log-slope-per-visit; (3): GHIV with county-fuel IVs and log-slope-per-visit and inflation-adjusted Medicare prices; (4): GHIV with log-slope-per-visit and inflation-adjusted Medicare prices. The demand model is estimated using 2-step nonlinear GMM. Robust standard errors are in parentheses. Estimates of county fixed effects are suppressed.

B Additional summary statistics

B.1 Length of stay

Length of stay	% of patients
0-7 days	30.89
8-30 days	29.38
31-90 days	21.7
91-179 days	9.52
180+ days	8.51

Table 12: Distribution of patient length-of-stay.

In my dataset, I observe the number of patients in a hospice who stayed for i) 0-7 days, ii) 8-30 days, iii) 31-90 days, iv) 91-179 days, v) 180+ days. Table 12 uses this data to construct the total number of patients in my dataset within each LOS bracket. Over 80% of the patients stay for less than 90 days.

Table 13 shows the distribution of people within each LOS-bracket by hospice-year. For nearly all hospices, most of their patients stay for less than 90 days.

	10%	25%	50%	75%	90%
0-7 days	19.76	24.66	30.3	36.31	41.81
8-30 days	21.93	25.05	29.29	33.01	36.33
31-90 days	16.15	18.9	21.7	25.17	29.44
91-179 days	5.45	7.47	9.41	11.43	13.92
180 days	2.24	4.48	7.2	10.73	14.52

Table 13: Distribution of shares of lengths-of-stay of patients. Each observation is at the hospice-year level.

B.2 Disease category

Diagnosis	% of patients
Cancer	39.6
Heart	10.27
Dementia	13.57
Lung	6.39
Kidney	2.49
Brain stroke	4.47

Table 14: Percentage of hospice patients by diagnosis.

The dataset classifies the patients at a hospice by broad categories of diagnosis. This allows me to see which diagnosis are more prevalent and whether hospices specialize in certain diagnosis over others. Table 14 shows that most hospice patients suffer from cancer; the second and third largest diagnosis categories are heart disease and dementia.

To see if hospices are specializing by diagnosis, or whether patients select into hospices by their diagnosis, Table 15 shows the distribution of patients of each diagnosis by hospice-year. This shows that most of a hospice's patients are suffering from cancer, heart disease or dementia, and there is no clear sign that a hospice gets most of its patients from one disease category alone.

	10%	25%	50%	75%	90%
Cancer	24.23	31.25	39.63	49.42	60.0
Non cancer	40.0	50.52	60.36	68.75	75.73
Heart disease	3.91	6.67	9.81	13.29	17.82
Dementia	3.33	6.5	11.72	18.03	25.0
Lung disease	2.61	4.27	6.18	8.46	11.12
Kidney	0.0	1.34	2.32	3.49	5.21
Liver	0.0	0.98	1.85	2.91	4.3
Brain stroke	0.0	1.53	3.48	5.89	8.96
HIV	0.0	0.0	0.0	0.13	0.46
Coma	0.0	0.0	0.0	0.0	0.35
Diabetes	0.0	0.0	0.0	0.0	0.44
ALS	0.0	0.0	0.26	0.61	1.03

Table 15: Distribution of shares of patients at a hospice per disease category. Each observation is at the hospice-year level.

B.3 Care type

	10%	25%	50%	75%	90%
Routine care	98.72	99.46	99.8	99.94	100.0
Inpatient care	0.0	0.0	0.04	0.23	0.69
Respite care	0.0	0.0	0.06	0.16	0.36
Continuous care	0.0	0.0	0.0	0.01	0.08

Table 16: Distribution of days of service provided for each hospice-year per care-type.

Medicare reimburses hospices with rates based on whether it provided routine care, inpatient care, respite care, and continuous care. Routine care has the lowest Medicare reimbursement rate, while other care types are much higher (see Figure 7).

The dataset divides the total days of care provided by a hospice into days providing each type of care. I calculate the fraction of care days in my dataset for each type of care.

Table 16 shows that nearly all of a hospice's care days are for routine care.

	10%	25%	50%	75%	90%
Registered nurses	8.82	11.42	14.44	18.21	22.48
Licensed vocational nurses	0.0	0.0	0.64	4.43	8.9
Homemakers	5.39	7.69	11.24	16.85	24.21
Physicians	0.0	0.0	0.01	0.25	0.77
Social service workers	1.74	2.59	3.55	4.73	6.05
Chaplains	0.27	1.02	1.84	2.87	4.08

Table 17: Distribution of visits by staff type for each hospice-year. The unit is visit per patient.

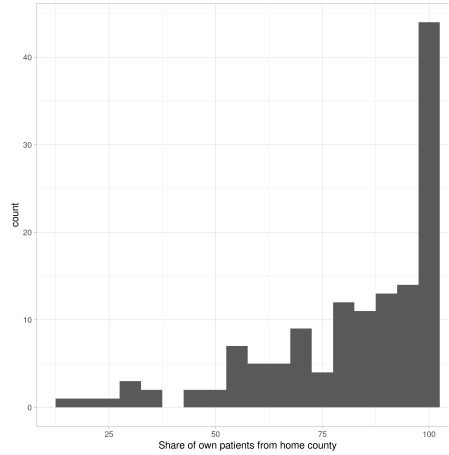


Figure 5: Histogram of home county share - percentage of a hospice’s patients from the county where it is located - for hospices in 2014.

B.4 Market definition

While I define my market to be at the level of a county, some papers on healthcare providers define markets at a larger level. Furthermore, some hospices advertise as being available for service in multiple counties at once.

In my setting, I can rule out larger market definitions using my dataset. For each hospice, the dataset includes measures of the fraction of patients that arrive from each county. This allows me to isolate the fraction of a hospice’s patients coming from the “home county” of the hospice (i.e. the county where the hospice is located) versus the fraction that is coming from an “away county”.

The distribution of home county share for each hospice in 2014 is shown in Figure 5. As this and Table 18 make clear, the vast majority of a hospice’s patients arrive from the county where it is located. As a result, I define the relevant market to be at the county level.

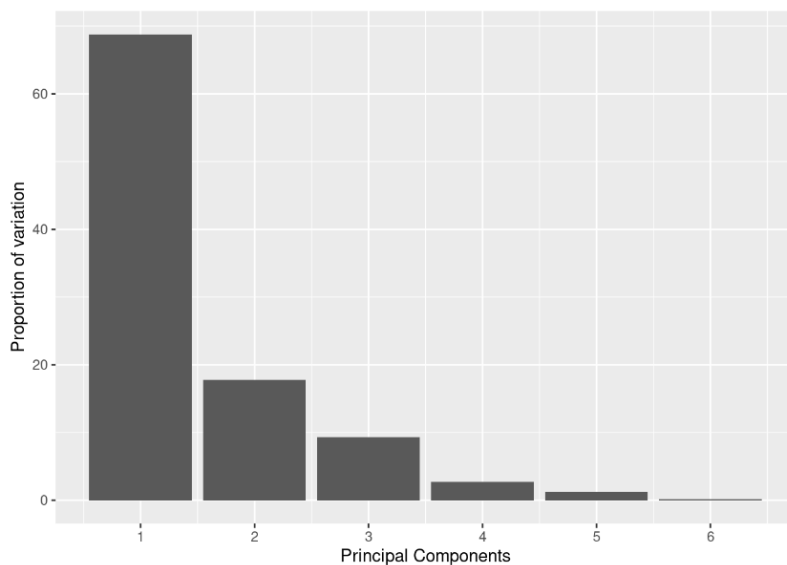
I focus on the following 28 counties in California: Alameda, Contra Costa, Amador, Butte, San Joaquin, El Dorado, Fresno, Humboldt, Del Norte, Imperial, Kern, Kings, Lake,

	25%	50%	75%
Home county share	70.6	88.4	99.0

Table 18: Distribution of days of service provided for each hospice-year per care-type.

Madera, Marin, Mariposa, Mendocino, Merced, Monterey, Napa, Nevada, Placer, Sacramento, San Francisco, San Luis Obispo, San Mateo, Santa Barbara, Santa Clara, Santa Cruz, Shasta, Siskiyou, Solano, Sonoma, Stanislaus, Sutter, Tehama, Tulare, Tuolumne, Yolo.

B.5 Principal components analysis of visits by staff type



	PC1	PC2	PC3	PC4	PC5	PC6
rmpat	0.4223153	0.7939506	-0.3565230	0.2495343	0.0433376	0.0061776
physnpat	0.0151012	-0.0068927	-0.0221099	-0.0032503	-0.0416916	-0.9987426
socsvcpat	0.1317572	0.1580743	-0.1338480	-0.9503556	0.1912078	-0.0010246
lvnpat	0.1834381	-0.5006202	-0.8411163	0.0710564	0.0519645	0.0224485
homemkrpat	0.8730719	-0.3053850	0.3769924	0.0254161	0.0411040	0.0051643
chaplainpat	0.0903201	0.0269610	-0.0699097	-0.1698567	-0.9774627	0.0440833

Figure 6: Principal components analysis of visits by staff type. The staff types are (in order of presentation) Registered Nurse, Physician, Social service worker, Licensed Vocational nurse, Home-maker, and Chaplains. The PCA analysis shows that the first Principal Component explains most of the variation in visits by a hospice.

C Additional Results

C.1 First stage of 2SLS

In this subsection I dig into the 2SLS regression reported in Table 5. The endogenous variable in that regression was the dynamic state “Average reputation of rivals - own reputation (lagged)”. The instruments used were ξ_{jt-1} and $X'_{jt-1}\beta_d$.

Below I disentangle the first stage regression for the 2SLS. Fundamentally, I want to make sure that the instruments are moving the dynamic state in accordance with economic theory. Briefly, a higher ξ_j or $X_j\beta_d$ should cause hospice j to choose lower quality and therefore get lower reputation. On the other hand the rivals would choose higher quality and therefore improve their reputation. These are insights I get from conducting counterfactuals involving heterogeneous hospices.

I therefore run the first-stage IV regression separately on two variables: Average reputation of rivals (lagged) and own reputation stock (lagged). The tables below show the regression of each variable on the exogenous variables and instruments. They confirm that the instruments move the dynamic states in the direction I expect.

Table 19: Estimates from first stage of 2SLS regression.

	Own reputation (lagged)
Count of rivals	0.014*** (0.002)
For-profit	0.061** (0.023)
Fuel price	0.017 (0.009)
Medicare rate	2.88×10^{-5} *** (2.372×10^{-6})
Average $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$	0.006 (0.018)
Average ξ_{jt} of rivals - own ξ_{jt}	0.084*** (0.016)
ξ_{jt-1}	-0.035** (0.011)
$X'_{jt-1}\beta_d$	-0.076 (0.055)
Firm Fixed Effects	Yes
Obs.	1,608

Notes: Estimates of hospice-specific fixed effects suppressed.

Table 20: Estimates from first stage of 2SLS regression.

	Average reputation of rivals (lagged)
Count of rivals	-0.005** (0.002)
For-profit	0.010 (0.018)
Fuel price	0.024** (0.007)
Medicare rate	1.91×10^{-5} *** (1.87×10^{-6})
Average $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$	-0.253*** (0.014)
Average ξ_{jt} of rivals - own ξ_{jt}	-0.062*** (0.013)
ξ_{jt-1}	0.009 (0.009)
$X'_{jt-1}\beta_d$	0.026 (0.043)
Firm Fixed Effects	Yes
Obs.	1,608

Notes: Estimates of hospice-specific fixed effects suppressed.

C.2 Hybrid scheme counterfactual with 3 firms

Table 21: Reimbursement schemes that lead to 0.76 visits-per-patient-day by three identical hospices.

Per-day rate	Per-visit rate	Medicare Spending (normalized)	Ex-Ante Value Function (normalized)
197.3	0.0	1.0	1.0
177.0	20.0	0.97	0.93
122.0	80.0	0.92	0.73
61.0	150.0	0.89	0.52

Notes: The rates are in US dollars. The total Medicare spending associated with each scheme is normalized with respect to the spending of the per-day scheme. The average ex-ante value function is normalized with respect to the ex-ante value function of the per-day scheme. As we move down rows, equilibrium quality stays the same yet Medicare spending declines.

D Additional data preparation for structural estimation

Censoring quality variable: To deal with high outlier values of the quality choice variable, I censor the visits-per-patient-day variable at the 90th percentile. That is, for quality values which are greater than the 90th percentile, I recode those values to be equal to the 90th percentile value.

Missing observations and fixing Firm IDs: I perform some additional data-cleaning before doing the analysis. First, for 24 hospices, it seems from looking at the data that a hospice forgot to respond in a certain year. This conjecture is due to the fact that the hospice was operating and serving a large number of patients in both the past few and next few years, but suddenly had its observation missing in one year. These non-responses are scattered across several years, with most in 2003 and only one after 2010. For these instances, I copy over the hospice's previous year's sales and characteristics into the missing year. This comprises 24 imputed data points in my dataset of approximately 2000 observations.

The data provider CHAI assigns an ID to each hospice, called the OSHPD ID. For 3 OSHPD IDs, they disappear from the data for 3-4 years before returning. I classify these as new entries by different entities.

Finally, for a few hospices, their OSHPD IDs change despite everything else about them staying the same (their name, location, etc.). For these hospices, I harmonize each to have the same OSHPD ID throughout.

I am happy to supply the code file for verification.

Imputing reputation stock of older hospices: For hospices which enter after 2002 (earliest period in my dataset), I can construct their exact reputation stock because I set their reputation during entry to be 0. However, a number of hospices are present from before the year 2002, and for them, I do not observe their pre-2002 quality choices.

While this does not affect my demand estimation (observe that the [Berry \(1994\)](#) allows me to difference out log of outside option share to account for rivals' characteristics), it will prevent me from conducting my supply side estimation. This is because a hospice's quality choice in each observed state is influenced by all its rivals' reputation stocks.

As a result, for the year 2002, I impute the reputation stock of hospices arriving before 2002 using a flexible functional form. First, for all hospices that arrived *after* 2002, I regress their reputation stocks on a large number of observed covariates. I use these regression coefficients to impute the reputation stocks – for the year 2002 – of all hospices that arrived before 2002. Now that I know their reputation stock coming into year 2002, I can use all their successive quality choices to update their reputation stock in future periods.

E Model estimation and solution

E.1 Demand model

For demand estimation, we estimate the following set of equations:

$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n \ln(s_{j|gt}) + \xi_{jt} + \eta[a_{jt} + (1 - \tau)a_{jt-1} + (1 - \tau)^2 a_{jt-2} + \dots]$$

$$\xi_{jt} = \rho \xi_{jt-1} + \varepsilon_{jt}$$

This model is estimated via 2-step nonlinear GMM (the nonlinearity stems from the fact that τ and η enter multiplicatively, and that ξ needs to be differenced out and ρ estimated). Let the matrix of instruments be Z . For the first step, I set the weighting matrix to be the inverse of $Z'Z$. For the second step, I construct the optimal weighting matrix using the residuals from the first-step GMM.

The 2-step GMM method requires estimating a large number of parameters, including county fixed effects and coefficients on hospice characteristics. A trick to easing computational burden is to note that only τ and ρ enter nonlinearly, so I can concentrate out all the linear parameters and minimize over (τ, ρ) .

E.2 Supply estimation via [Bajari et al. \(2007\)](#)

E.2.1 How market share is calculated from demand estimates

To calculate profit, I need to calculate market shares of all firms for a given state $\mathbf{x}_m$. This is done using the analytical formulae for nested logit (see [Berry \(1994\)](#)). Let

$$\delta_{jt} = \alpha_m + X'_{jt}\beta_d + \sigma_n \ln(s_{j|gt}) + \psi_{jt} + \xi_{jt}$$

where σ_n is the nest parameter that captures correlated preferences within the nest. Then the market share can be written as:

$$s_j(\delta, \sigma_n) = \bar{s}_{j/g}(s, \sigma_n) \bar{d}_g(\delta, \sigma_n) = \frac{e^{\delta/(1-\sigma_n)}}{D_g^{\sigma_n} \left[\sum_g D_g^{(1-\sigma_n)} \right]}$$

where

$$D_g \equiv \sum_{j \in G_g} e^{\delta_j/(1-\sigma_n)}$$

E.2.2 Equilibrium concepts

Markov Perfect Equilibrium: Markov Perfect Equilibrium is a refinement of Nash Equilibrium that is a type of Subgame Perfect Equilibrium. It says that a player's strategy depends only on "payoff-relevant" state variables rather than on potentially the whole history of play. Intuitively, it limits the number of variables that a player looks at before taking an action. This introduces simplicity as well as ruling out a vast number of multiple equilibria. See [Ericson and Pakes \(1995\)](#) for a detailed discussion.

Symmetric and anonymous policy functions: I also impose the policy functions to be symmetric and anonymous. The symmetry assumption imposes that all firms have the same policy function. The anonymity assumption imposes that a firm does not condition its choice on the identity of its rivals.

E.2.3 Forward simulation and optimization

A brief overview of the algorithm is as follows. Fix a guess of structural parameters θ . For every observed state $\mathbf{x}_t$ and hospice j (i.e. each observation in my hospice-level panel data), I estimate the choice-specific value function for each quality level. For quality choice a_j by hospice j , a single simulated path is constructed as follows:

1. The first-stage policy function is used to predict quality choices by all firms, conditional on the simulated market state in that period.
2. The reputation stock of each firm is updated based on its quality choice.
3. Market share of each hospice is realized. Each incumbent receives a payoff.
4. The entry transition function is used to predict the number of potential entrants who apply for entry. These entries are implemented in the next period.
5. The empirical exit probabilities are used to predict whether any incumbent exits.
6. Using the state variable transition functions from first-stage, all state variables update. This includes unobserved demand shocks, market size, and Medicare rates.
7. The game moves to the next period, and I repeat this algorithm.

The above simulation is run for many periods until the discounted profit is driven close to zero; in my estimation I set this to be 80 periods. The simulations use Halton sequences. Summing together the discounted per-period payoffs gives me the total payoff from hospice j choosing quality a_j for a single simulated path. I repeat this simulation 200 times and average over the total payoffs. This gives me the choice-specific value function of hospice j choosing quality a_j in state $\mathbf{x}_t$, $\hat{v}_j(a_j, \mathbf{x}_t, \sigma^*; \theta)$. In short, I average over the sum of discounted payoffs for each simulation run, and this gives me the choice-specific value function for that action-state combination.

The choice-specific value functions are constructed for every quality $a \in \mathcal{A}$. These are then used to construct model-predicted choice probabilities $\Psi_j(a_j, \mathbf{x}_t, \sigma^*; \theta)$ from the

structural model using Equation (9). The model-predicted choice probabilities are interpreted as follows – given state $\mathbf{x}_t$, how likely is it that hospice j would choose action a_j , conditional on our guess of structural parameters θ .

Next, I describe how these model-predicted choice probabilities are used to estimate the hospice cost function. The intuition is that I search for values of parameters θ that minimize the distance between model-predicted choice probabilities and observed choices.

The dynamic oligopoly model is estimated using two-step GMM. The GMM residual term for observation n is:

$$\Xi_n(\theta) = a_n^{data} - \sum_{a_n \in \mathcal{A}} a_n \hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$$

where $\hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$ is the predicted choice probability for action a_n , a_n^{data} is the observed quality choice for observation n , θ is the guess of structural parameters, and $\mathbf{x}_n^{data}$ is the market state for observation n . The resulting GMM optimization problem is:

$$\min_{\theta} \left[\frac{1}{N} \sum_n Z_n' \Xi_n(\theta) \right]' \hat{W} \left[\frac{1}{N} \sum_n Z_n' \Xi_n(\theta) \right]$$

This is solved with 2-step GMM. The instruments used in GMM are the county indicators and the variables in the first-stage empirical policy functions (with the exception of the firm fixed effects), giving us the moment conditions $E[Z' \Xi_n(\theta)]$.

Let the matrix of instruments be Z . For the first step, I set the weighting matrix to be the inverse of $Z'Z$. For the second step, I construct the optimal weighting matrix using the residuals from the first-step GMM.

A way I ease computational burden is by using the “linear-in-parameters” method outlined in [Bajari et al. \(2007\)](#) and [Ryan \(2012\)](#). Similar to their setup, the hospice payoff functions can be split up in such a way that I do not need to redo the forward simulation for each guess of the cost parameters. Instead, I can perform the forward simulation just once and then use the trial cost parameters to rescale the choice-specific value functions and then evaluate the objective function. Please see [Bajari et al. \(2007\)](#) and [Ryan \(2012\)](#) for a careful explanation.

E.3 Counterfactual model solution via policy iteration

E.3.1 Policy iteration

To conduct counterfactuals I solve the model by policy iteration method. This requires solving a nested fixed-point problem; more specifically, I need to calculate $\Psi_j(a_j, \mathbf{x}_m, \sigma^*)$, $\hat{v}(a_j, \mathbf{x})$ and $V(\mathbf{x})$ such that Equations (14), (15), and (16) simultaneously hold:

$$\Psi_j(a_j, \mathbf{x}_m, \sigma^*) = \frac{\exp\{\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*)/\theta_e\}}{\sum_{a \in \mathcal{A}} \exp\{\hat{v}_j(a, \mathbf{x}_m, \sigma^*)/\theta_e\}} \quad (14)$$

$$\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*) = \sum_{\mathbf{a}_{-j} \in \mathcal{A}_{-j}} \left\{ \left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) + \beta V(\mathbf{x}'_m, \sigma^*) \right] F(\mathbf{x}'_m | \mathbf{x}_m, a_j, \mathbf{a}_{-j}) \prod_{n \neq j} \Psi_n(\mathbf{a}_{-j}[n], \mathbf{x}_m, \sigma^*) \right\} \quad (15)$$

where $\Psi_n(\cdot, \mathbf{x}_m, \sigma^*)$ is the CCP of firm n , $\mathcal{A}_{-j}$ is the set of all possible rival action profile vectors, and $\mathbf{a}_{-j}[n]$ is the action by firm n .

$$V_j(\mathbf{x}_m, \sigma^*) = \theta_e \left[0.577216 + \ln \left(\sum_{a_j \in \mathcal{A}} e^{\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*)/\theta_e} \right) \right] \quad (16)$$

I solve the model by iterating between the three equations until convergence. Policy iteration proceeds as follows. I iterate between Equations (15) and (16) several times, then iterate between Equations (14) and (15) several times. I repeat this until successive CCPs and ex-ante value functions are within some pre-determined tolerance.

F Additional discussion on interpreting the structural model and counterfactuals

F.1 Interpreting entry-exit incentives from ex-ante value functions

I report the ex-ante value functions for my counterfactuals to show how entry-exit incentives change. The overall intuition is that as market conditions improve, potential entrants will choose to enter; as market conditions worsen, incumbents will choose to exit.

I now give two examples to elaborate.

1. In a sequential entry game of homogeneous firms with no private shocks (e.g. [Alam and Conti \(2024\)](#) and [Igami \(2018\)](#)), the equilibrium number of firms n^* is given by:

$$V(n^*, s_0) \geq \kappa_{entry}$$

where $V(\cdot, \cdot)$ is the value function, s_0 is the starting state, and κ_{entry} is the entry cost. We can see here how a policy change that uniformly raises $V(\cdot, \cdot)$ will raise n^* , and the opposite will lower n^* .

2. In dynamic games studying exit, the exit decision compares the value function from staying on with the scrap value, i.e. $V(x_m) \leq \phi_{scrap}$. If this inequality holds, then subject to logit shocks, the firm is likely to exit. Any policy that lowers $V(\cdot)$ therefore makes exit more likely.

While ex-ante value functions are indicative of how entry-exit incentives change, they do not track them exactly in my setting; this is because there are economic mechanisms I rule out by not explicitly modeling entry and exit decisions. For instance, if entry by new firms is a possibility, then incumbents may react by building up reputation to deter entry. If exit were a possibility, then incumbents will account for the possibility of themselves or their rivals exiting when making quality choice today. I do not allow for these in my counterfactual model solution.

I do not model entry and exit explicitly for several reasons. First, entry is infrequent and exit is rare in my data. Second, it will increase the computational burden of solving

my counterfactuals (both through greater computational need as well as increasing the state space size). Finally, I don't find evidence of entry deterrence through reputation accumulation in my data, and so do not want to include it as an economic mechanism when doing the counterfactuals.

F.2 Interpreting the profit function

Let the profit from patient i be

$$\Pi_i = P_m^{MCAR} LOS_i - \gamma_0 - \gamma_1 a_j LOS_i$$

This is a random variable, because LOS_i varies by patient i , and is unknown to the hospice when it chooses quality a_j or even during patient enrollment.

The expected profit per patient integrates over the above expression with respect to the distribution of LOS :

$$\mathbb{E}_{LOS}[\Pi_i] = P_m^{MCAR} \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}$$

where $\mathbb{E}_{LOS}[LOS_i] \equiv \overline{LOS}$ to match the notation in the main text.

Using this notation, the profit function used in the paper is therefore:

$$\begin{aligned} \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) &= -\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \mathbb{E}_{LOS}[\Pi_i] \\ \implies \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) &= -\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} P_m^{MCAR} \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS} \\ \implies \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) &= -\phi_j + M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) [P_m^{MCAR} \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}] \end{aligned}$$

which is the profit function reported in the main text.

Note that this is different from the profit function being the sum of *realized* profits per patient:

$$\tilde{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + \left\{ \sum_{i=1}^{M_{msj}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}$$

The expectation of this profit function lines up with the sum of expected profits:

$$\mathbb{E}_{LOS}[-\phi_j + \left\{ \sum_{i=1}^{M_{msj}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}] = -\phi_j + M_{msj}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) [P_m^{MCAR} \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}]$$

However, it also exhibits variance.

$$\mathbb{V}_{LOS}[-\phi_j + \left\{ \sum_{i=1}^{M_{msj}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}] \neq 0$$

while the sum of expected profits does not

$$\mathbb{V}_{LOS}[-\phi_j + \sum_{i=1}^{M_{msj}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \mathbb{E}_{LOS}[\Pi_i]] = 0$$

G Relevant webpages

G.1 Links to hospice quality measurement

In recent years, CMS and AHRQ have combined multiple measures from several data sources to construct quality indices for hospices. The following links to their websites:

1. <https://www.cms.gov/medicare/quality/hospice/current-measures>
2. <https://www.ahrq.gov/talkingquality/measures/setting/long-term-care/hospice.html>

This shows that the agencies are tracking a variety of measures to approximate quality. I do not have access to this data, and furthermore many of these were not available over the

entire period of my data (2002-2018). Instead, my claim is that visits-per-patient-day is a good proxy for the effort exerted by the hospice, since visits are usually the most costly part of care.

Here is a quote from a New York Times article (<https://www.nytimes.com/2023/06/10/health/hospice-profits.html>) from a former hospice admin that backs up my claim.

“In the nearly 20 years that Megan Stainer worked in nursing homes in and around Detroit, she could almost always tell which patients near death were receiving care from nonprofit hospice organizations and which from for-profit hospices.

“There were really stark differences,” said Ms. Stainer, 45, a licensed practical nurse.

Looking at their medical charts, “the nonprofit patients always had the most visits: nurses, chaplains, social workers. The nonprofit hospices responded quickly when the nursing home staff requested supplies and equipment.”

G.2 Quotes on hospice reputation

1. Some websites which detail my claim on how hospices are chosen can be found in the following:

- <https://www.vitas.com/hospice-and-palliative-care-basics/when-is-it-time-for-hospice/how-to-choose-a-hospice-provider>
- <https://www.cancer.org/treatment/end-of-life-care/hospice-care/how-to-find.html>
- <https://www.health.harvard.edu/staying-healthy/choosing-hospice>
- <https://hospicefoundation.org/End-of-Life-Support-and-Resources/Coping-with-Terminal-Illness/How-to-Choose>
- <https://rainbowhospice.org/hospice-care/choosing-hospice/>

2. Some quotes on how reputation should be considered when choosing a hospice are as follows.

- AmericanHospice.org: What do others say about this hospice? Get references both from people you know and from people in the field – e.g., local hospitals, nursing homes, clinicians. Ask anyone that you have connections to if they have had experience with the hospice and what their impressions are. Geriatric care managers can be a particularly good resource, as they often make referrals to hospices and hear from families about the care that was provided... How long has the hospice been in operation? If it has been around for a while, that's an indication of stability."
- HospiceFoundation.org: "Seek professional opinions. Ask clinicians, professional caregivers at nursing homes, geriatric care managers, or end-of-life doulas about their experience with a hospice. Talk to friends, family, and neighbors who have used hospice services and get their opinions about the experience with a provider."
- Vitas.com: "Evaluate the hospice provider's history and reputation before you decide. How long has it been in business? ... What do other patients or families say about their experiences?"
- Caringinfo.org: "Most hospice programs use family satisfaction surveys to obtain feedback about their services so they can make improvements. Ask the hospice to share a summary of their family satisfaction scores for the last several months with you. You can also ask to see their latest state or Medicare inspection report to see if there are care provision problems. Finally, you could ask to see the hospice provider's list of complaints from the past 12 months."

G.3 Example of a hospice reimbursement scheme

LOCATION BY COUNTIES	ROUTINE HOME CARE DAYS 1-60	ROUTINE HOME CARE DAYS 61+	SERVICE INTENSITY ADD-ON (HOURLY)	CONTINUOUS CARE (HOURLY)	INPATIENT RESPITE CARE	GENERAL INPATIENT CARE
	Revenue Code 0650	Revenue Code 0659	Revenue Code 0552	Revenue Code 0652	Revenue Code 0655	Revenue Code 0656
NATIONAL RATES	\$ 196.50	\$ 154.41	\$ 41.57	\$ 41.57	\$ 185.27	\$ 758.07
RURAL AREAS	\$ 237.03	\$ 186.26	\$ 50.15	\$ 50.15	\$ 215.38	\$ 903.74
ALAMEDA & CONTRA COSTA	\$ 294.39	\$ 231.33	\$ 62.28	\$ 62.28	\$ 257.98	\$ 1,109.87
BUTTE	\$ 214.74	\$ 168.74	\$ 45.43	\$ 45.43	\$ 198.82	\$ 823.63
FRESNO	\$ 208.62	\$ 163.94	\$ 44.14	\$ 44.14	\$ 194.28	\$ 801.64
IMPERIAL	\$ 180.42	\$ 141.77	\$ 38.17	\$ 38.17	\$ 173.33	\$ 700.28
KERN	\$ 223.41	\$ 175.56	\$ 47.26	\$ 47.26	\$ 205.26	\$ 854.78
KINGS	\$ 206.61	\$ 162.36	\$ 43.71	\$ 43.71	\$ 192.78	\$ 794.41
LOS ANGELES	\$ 234.05	\$ 183.92	\$ 49.52	\$ 49.52	\$ 213.16	\$ 893.02

Figure 7: Snapshot of Medicare reimbursement scheme in 2018. The reimbursement rate varies by county based on its Medicare wage index, and varies by the type of care provided.

H Hospice cost data

Per-period fixed cost:

- Sums: General service cost centers, Other Hospice Service Costs, Bereavement program costs, Other program costs, Writeoffs.
- Subtracts: Revenue from managed care, Revenue from private insurance, Revenue from self-pay, Revenue from other payers, Grants, Donations/Contributions, Unrelated Business Income, Total other operating revenue.

Non-visit cost:

- Sums: Inpatient care service expense (both general and respite care), Radiation therapy cost, Chemotherapy cost, Program supervision cost, Volunteer program costs.
- This is divided by total number of patients in that year to get the non-visit cost per patient (“intercept cost”).

Slope cost:

- Drugs/biologicals/infusion cost, Durable medical equipment/oxygen cost, Patient transportation cost, Imaging/lab/diagnostics cost, Medical supplies cost, Outpatient services cost, Visiting services cost (all staff types).
- This is divided by total number of visits in that year to get the cost of an additional visit (“slope cost”).

Figure 8: List of variables in the hospice cost data.

DETAIL OF OPERATING EXPENSES

(do not input "\$" signs, commas, or decimals; round up to whole dollar)

Use data from Medicare Cost Report where applicable.

Line No.		Total (1)
	General Service Cost Centers	
30	Administrative and General	
	Inpatient Care Service	
31	Inpatient – General Care	
32	Inpatient – Respite Care	
	Program Supervision	
35	Hospice Program / Team Supervision (Non-Visit Wages)	
	Visiting Services	
36	Physician Services	
37	Nursing Care	
38	Rehabilitation Services (PT, OT, Speech)	
39	Medical Social Services – Direct	
40	Spiritual Counseling	
41	Dietary Counseling	
42	Counseling – Other	
43	Home Health Aides and Homemakers	
44	Other Visiting Services	
	Hospice Service Cost Centers	
45	Drugs, Biologicals, and Infusion	
46	Durable Medical Equipment / Oxygen	
47	Patient Transportation	
48	Imaging, Lab, and Diagnostics	
49	Medical Supplies	
50	Outpatient Services (including ER Dept.)	
51	Radiation Therapy	
52	Chemotherapy	
53	Other Hospice Service Costs	
	Other Hospice Costs	
54	Bereavement Program Costs	
55	Volunteer Program Costs	
56	Fundraising Costs	
	Other Costs	
57	Other Program Costs*	
59	Total Operating Expenses	

* Program costs including community education and outreach program costs.

Table 22: Regression of log of per-period fixed cost, log of intercept cost, and slope cost (from cost data) on county dummies.

	(1)	(2)	(3)
Alameda	13.920*** (0.111)	5.262*** (0.157)	111.949 (102.670)
Amador	12.306*** (0.516)	6.163*** (0.732)	132.778 (298.813)
Butte	13.109*** (0.214)	5.609*** (0.303)	118.234 (167.659)
Contra Costa	13.718*** (0.133)	5.460*** (0.188)	141.194 (122.592)
Del Norte	10.812*** (1.264)	5.493*** (1.792)	200.617 (871.181)
El Dorado	13.447*** (0.298)	5.049*** (0.422)	128.679 (275.492)
Fresno	13.043*** (0.140)	6.028*** (0.198)	114.276 (129.868)
Humboldt	13.900*** (0.307)	5.157*** (0.435)	119.419 (298.813)
Imperial	12.814*** (0.478)	7.155*** (0.677)	62.458 (341.705)
Kern	12.943*** (0.132)	5.717*** (0.187)	370.076*** (120.234)
Kings	13.280*** (0.351)	5.861*** (0.497)	316.966 (298.813)
Lake	12.594*** (0.316)	5.407*** (0.448)	110.496 (308.009)
Madera	9.335*** (1.264)	4.375** (1.792)	132.881 (550.983)
Marin	13.708*** (0.283)	6.830*** (0.401)	130.809 (251.488)
Mariposa	11.003*** (0.381)	6.630*** (0.540)	115.431 (275.492)
Mendocino	12.982*** (0.316)	6.192*** (0.448)	151.994 (298.813)
Merced	12.737*** (0.243)	5.164*** (0.345)	100.129 (202.545)
Monterey	13.732*** (0.202)	6.021*** (0.287)	177.939 (177.829)
Napa	13.257*** (0.316)	5.133*** (0.448)	116.208 (298.813)
Nevada	12.922*** (0.248)	6.513*** (0.351)	186.773 (211.292)
Placer	13.732*** (0.184)	6.263*** (0.261)	160.865 (177.829)
Sacramento	13.731*** (0.117)	5.434*** (0.166)	128.329 (107.643)
San Francisco	13.117*** (0.167)	6.368*** (0.237)	200.572 (145.197)
San Joaquin	12.475*** (0.174)	6.288*** (0.246)	110.236 (154.004)
San Luis Obispo	13.943*** (0.217)	5.847*** (0.307)	109.273 (205.339)
San Mateo	14.021*** (0.156)	6.485*** (0.221)	141.582 (145.197)
Santa Barbara	13.720*** (0.170)	5.944*** (0.242)	108.140 (155.222)
Santa Clara	14.171*** (0.124)	5.493*** (0.176)	122.940 (113.902)
Santa Cruz	14.012*** (0.276)	3.892*** (0.391)	139.707 (246.407)
Shasta	12.472*** (0.269)	4.536*** (0.382)	139.374 (197.284)
Siskiyou	11.882*** (0.220)	6.034*** (0.312)	1143.690*** (185.736)
Solano	13.837*** (0.243)	5.298*** (0.345)	173.224 (232.833)
Sonoma	13.998*** (0.152)	4.554*** (0.216)	145.786 (126.404)
Stanislaus	13.695*** (0.193)	6.583*** (0.273)	118.777 (174.236)
Sutter	13.377*** (0.365)	6.824*** (0.517)	139.874 (262.671)
Tehama	12.896*** (0.447)	2.542*** (0.634)	163.460 (308.009)
Tulare	13.075*** (0.264)	5.524*** (0.374)	170.015 (194.802)
Tuolumne	12.959*** (0.338)	5.716*** (0.479)	167.617 (308.009)
Yolo	13.262*** (0.316)	5.083*** (0.448)	128.558 (290.394)
Yuba	12.420*** (0.381)	6.142*** (0.540)	206.696 (355.658)
N	1,495	1,495	1,806
R ²	0.991	0.913	0.038

I Additional discussion on institutional details and modeling decisions

How does advertising play a role here? Do hospices spend on advertising to build up name recognition?

I allow for unobserved demand shocks in my demand model, and advertising gets picked up there. Furthermore, my understanding is that advertising was pretty low in the hospice industry during my data period; in fact, hospices have generally suffered from lack of awareness. I don't model advertising as a strategic choice variable in my supply estimation.

What is the variation in your data that lets you identify i) your key demand parameters, ii) your key cost parameters?

In a simplified sense, reputation parameters are identified by regressing current market share on a function of current and past quality choices. The nonlinear regression tells me how much impact past choices have on current share, as well as how important the total reputation stock is for explaining current market share.

The cost parameters are identified through a revealed preference argument. Let's think about the cost of increasing quality from 0.44 to 0.56. Accounting for the demand system and rivals, an increase in quality will lead to higher market share. So the main reason I would observe a firm choosing 0.44 instead of 0.56 is if the increase in total revenue is more than offset by the corresponding increase in cost.

Are there capacity constraints for hospices, and can those be driving the results of the demand estimation? (Alternatively: every hospice has a fixed number of nurses. When there are more hospices, each hospice has fewer patients, so they make more visits per patient. As more firms come in there are greater visits per patient. Then as firms exit, number of visits per patient declines.)

Several answers to this:

1. I see total number of visits by a hospice increases as there are more competitors.

This goes against the idea that hospices have a fixed number of staff as competitors enter; instead, it suggests that hospices respond to entry by making more visits in total.

2. Capacity constraint is unlikely to be an issue. Hospices not really limited by physical space since they generally serve patients at patient residences. The way it works with hospice hiring is: hospices hire nurses around the area where they operate. The hospice nurses often drive directly from their home to various patient residences. Therefore, there isn't a capacity constraint with hospices like there is with nursing homes or hospitals.
3. While capacity constraint can create some correlation between past visits-per-patient and current market share, it would have the opposite effect to what I see. I find that market share is *positively* correlated with visits-per-patient, which suggests the opposite.

How does market clearing work in this setting?

The equilibrating mechanism here is reputation. Demand increases in reputation, supply decreases in reputation because reputation is costly to build. Simplest case is to think of homogeneous hospices with no entry-exit. The equilibrium reputation level would be the one that sets demand equal to supply.

As an example: If demand exceeds supply, equilibrium reputation declines until demand equals supply. As reputation declines, consumers switch to the outside option as well.

In my model, hospices are differentiated along other dimensions, so there isn't a single reputation level at which the market clears.

So how does this relate to the Markov Perfect Equilibrium? The MPE restricts the state variables that can enter a firm's policy function, so by itself it doesn't say much about the market clearing equilibrium behavior. One way to think is - the derived policy function is an equilibrium policy function. It says, for given values of every state variables, what choice a firm will make such that it will be the Nash equilibrium choice. Thus, the MPE policy function gives us the Nash equilibrium of quality choices among hospices. When

forming this policy function the hospices take into account the demand function, so the market clearing conditions are satisfied in the background.

Are you worried about patient selection issues by hospices? Could there be some sort of selective admissions going on? Maybe hospices avoid certain types of patients, or higher quality hospices get more severe patients, or hospices which are making more visits have more severe patients.

Two possible types of selection:

1. High-severity patients selecting into (maybe high quality) hospices: Spoke to people familiar with the industry and read up on industry profiles, this does not seem to be happening. All patients die by the end of their length-of-stay, so they are all in bad conditions to begin with. Furthermore, all patients would presumably want high visits so that they are taken care of, since hospice care is free for them.
2. Hospices selecting low-severity patients: Again, from speaking to people familiar with the industry and reading up on industry profiles, this does not seem to be happening. You can also imagine that the hospice can simply enroll a high-severity patient and just not give them many visits, since the hospice decides how many visits to make. I also haven't heard of hospices turning patients away because they're displaying severe symptoms.

In terms of observable patient characteristics such as diagnosis there doesn't seem to be any clear indication of selection in my data.

Any differences on length-of-stay by patient pool won't pose a problem, because my measure of quality is visit-per-patient-day. For-profits and non-profits seem to have different lengths-of-stay on average; since my quality measure is visit-per-patient-day, any such differences in average length-of-stay are accounted for.

I'm not saying that this sort of selection does not happen at all, but it should be a good approximation in this setting.

How do consumers *know* reputation? How can consumers observe quality, or do consumers need to perfectly observe past quality choices?

All consumers need is to be able to observe the reputation index of each hospice. This reputation index can be formed within the community from past experiences with a

hospice. The consumer can find out this reputation index by talking to members of the community such as the social worker, physician, support groups, etc. It doesn't require an individual consumer to have carefully kept track of every quality choice made by a hospice in the past.